

# Schwarz Coupling for Solid Dynamics in Norma: Transmission Conditions and Interface Energy

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## Abstract

This note is the reference for the Schwarz transmission conditions implemented in Norma and for the measurements of their energy behavior in dynamic solid mechanics. It describes the conditions as they are implemented now, states the requirements that an energy-consistent coupling must satisfy, and summarizes the measurements on two benchmarks: a cantilever beam released from a bent shape, with nonconforming meshes at a flat interface, and nested cylinders of three materials launched against a rigid surface, with a closed curved interface. The findings are: Dirichlet transmission on nonconforming meshes injects energy and ends in element inversion on both benchmarks; the classical Robin–Robin condition, which has no absorbing term, does the same, usually sooner, at either of two Robin parameters; the impedance condition on an overlap boundary is exact on conforming meshes but cannot be made energy consistent on nonconforming ones, because the two sides exchange data on two different surfaces through operators that are not adjoint; and the adjoint-paired impedance condition on a shared interface, which transfers a single d’Alembert interface reaction through adjoint operators built from one cross-mass matrix, never injects energy on either benchmark, is exact at mild mesh contrast, and dissipates at strong contrast at a rate set by the part of the interface velocity jump that the two trace spaces cannot represent in common. The note records the explicit–explicit and different-time-step variants of that coupling, the behavior of the relaxation and Aitken acceleration of the Schwarz iteration, and the open questions, chiefly the stalled interface jump on the cylinders and the dependence of the converged solution on the Robin parameter. A closing section states the constrained formulation of the coupled problem that conserves energy by construction, from Gravouil and Combescure and from Prakash and Hjelmstad, locates the five points at which the nonoverlap Dirichlet–Neumann path departs from it, and gives the changes and the reference experiments by which Norma is to be brought into it.

*This document supersedes the July–August 2026 version of this note and the three earlier notes it consolidated. Measurements that motivated design decisions but concern superseded variants are kept only in summary.*

## Contents

|          |                                                     |          |
|----------|-----------------------------------------------------|----------|
| <b>1</b> | <b>Setting</b>                                      | <b>2</b> |
| 1.1      | Decompositions and notation . . . . .               | 2        |
| 1.2      | Vocabulary of the multi-domain controller . . . . . | 3        |
| 1.3      | Benchmarks . . . . .                                | 3        |
| 1.4      | Energy metrics . . . . .                            | 4        |

|          |                                                                         |           |
|----------|-------------------------------------------------------------------------|-----------|
| <b>2</b> | <b>Transmission conditions on a shared interface</b>                    | <b>5</b>  |
| 2.1      | Robin coupling . . . . .                                                | 5         |
| 2.2      | Impedance coupling . . . . .                                            | 5         |
| 2.3      | Reflection at the interface . . . . .                                   | 6         |
| <b>3</b> | <b>Transmission conditions on an overlap boundary</b>                   | <b>6</b>  |
| 3.1      | Dirichlet coupling . . . . .                                            | 6         |
| 3.2      | Impedance coupling . . . . .                                            | 6         |
| 3.3      | Summary of the four conditions . . . . .                                | 7         |
| <b>4</b> | <b>Requirements for energy consistency</b>                              | <b>8</b>  |
| <b>5</b> | <b>The adjoint-paired nonoverlap impedance coupling</b>                 | <b>9</b>  |
| 5.1      | Construction . . . . .                                                  | 9         |
| 5.2      | Explicit subdomains . . . . .                                           | 10        |
| 5.3      | Different time steps per subdomain . . . . .                            | 10        |
| 5.4      | Relaxation and Aitken acceleration . . . . .                            | 11        |
| <b>6</b> | <b>Measurements on the cantilever beam</b>                              | <b>12</b> |
| 6.1      | Overlap, conforming meshes: the partner traction . . . . .              | 12        |
| 6.2      | Overlap, nonconforming meshes: sign-indefinite interface work . . . . . | 12        |
| 6.3      | Overlap: remedies tried on the interface . . . . .                      | 13        |
| 6.4      | Nonoverlap: legacy and paired coupling . . . . .                        | 14        |
| 6.5      | Windowed controller stops . . . . .                                     | 17        |
| 6.6      | The five couplings with the current code . . . . .                      | 17        |
| 6.7      | The classical Robin–Robin condition . . . . .                           | 17        |
| <b>7</b> | <b>Measurements on the nested cylinders</b>                             | <b>19</b> |
| <b>8</b> | <b>The constrained formulation of the coupled problem</b>               | <b>21</b> |
| 8.1      | What the multiplier formulations prove . . . . .                        | 21        |
| 8.2      | The pseudo-energy identity of the subcycled exchange . . . . .          | 22        |
| 8.3      | Where the Dirichlet–Neumann path departs from it . . . . .              | 23        |
| 8.4      | Changes and reference experiments . . . . .                             | 24        |
| 8.5      | Measurements of the constrained exchange . . . . .                      | 25        |
| <b>9</b> | <b>Conclusions</b>                                                      | <b>33</b> |
| 9.1      | Findings . . . . .                                                      | 33        |
| 9.2      | Recommendations . . . . .                                               | 34        |
| 9.3      | Open questions . . . . .                                                | 35        |

# 1 Setting

## 1.1 Decompositions and notation

A solid body  $\Omega \subset \mathbb{R}^3$  is decomposed into two subdomains  $\Omega_1$  and  $\Omega_2$  that are integrated independently and coupled by a Schwarz iteration. The Schwarz framework itself (time controller, subdomain solves, convergence criteria) follows Mota, Tezaur, and Phlipot [1], and its contact

variant Mota et al. [2]. This note concerns the transmission conditions imposed on the coupling boundaries and their energy behavior. Norma supports two geometric configurations.

**Nonoverlap.** The subdomains partition  $\Omega$  and share an interface  $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ . Let  $\mathbf{n}_k$  be the outward unit normal to  $\Omega_k$  on  $\Gamma$  (so  $\mathbf{n}_1 = -\mathbf{n}_2$ ) and  $\mathbf{t}_k = \boldsymbol{\sigma}_k \mathbf{n}_k$  the traction on the side of  $\Omega_k$ . The coupled solution must satisfy

$$\mathbf{u}_1 = \mathbf{u}_2 \quad (\text{kinematic continuity}), \quad (1)$$

$$\mathbf{t}_1 + \mathbf{t}_2 = \mathbf{0} \quad (\text{traction balance}). \quad (2)$$

The Dirichlet–Neumann method imposes (1) on one side and (2) on the other. The Robin and impedance conditions of Sec. 2 impose on each side a linear combination of displacement, velocity, and traction.

**Overlap.** The subdomains intersect. Each  $\Omega_k$  has a portion of its boundary,  $\Gamma_k \subset \Omega_j$  with  $j \neq k$ , in the interior of the partner. That boundary is not shared with  $\Omega_j$ , so no mutual traction is directly available there; the partner provides a well-defined interior state at every point of  $\Gamma_k$ , and the conditions of Sec. 3 are built from that state.

Throughout,  $\mathbf{W}_k = \int_{\Gamma_k} \mathbf{N}_k \mathbf{N}_k^\top d\Gamma$  is the boundary mass matrix of side  $k$ ,  $\mathbf{P}$  a kinematic transfer operator that maps partner data to the interface nodes of a side, the subscript  $j$  denotes the partner of side  $k$ , and  $\llbracket \mathbf{q} \rrbracket = \mathbf{q}_k - \mathbf{q}_j$  is the interface jump of a field  $\mathbf{q}$ , own side minus partner, on the coupling surface of side  $k$ . All interface operators are assembled once at initialization on the reference configuration with the facet Gauss rule of the receiving side set, and are not updated with the deformation.

## 1.2 Vocabulary of the multi-domain controller

The controller advances in *stops*. Within a stop each subdomain advances to the stop end in its own *substeps*, and the *Schwarz iteration* repeats a *pass* over the subdomains (one solve of each, with interface data exchanged between them) until the exchanged data converge; the stop is then committed. The exchanged datum may be relaxed between Schwarz iterations with a factor  $\theta \in (0, 1]$ , fixed or adapted by Aitken extrapolation (Sec. 5.4). A stop that spans several substeps is a *window* (Sec. 6.5). A mesh pair is *nested* when every coarse node coincides with a fine node. Mesh ratios are written 1: $r$  with  $r = h_{\text{fine}}/h_{\text{coarse}}$ .

## 1.3 Benchmarks

**Cantilever beam.** A linear elastic bar ( $E = 6.895$  GPa,  $\nu = 0.25$ ,  $\rho = 2768$  kg/m<sup>3</sup>; 254 mm long, 25.4 mm tall, 12.7 mm wide) is clamped at one end and released at  $t = 0$  from the parabolic bending displacement  $u_y = 0.3937 x^2$  (tip deflection 25.4 mm), with initial energy 1506 J. The subdomain time step is 1  $\mu$ s; horizons are 1 and 10 ms. Subdomain integrators are trapezoidal Newmark (implicit, I) and central difference (explicit, E), in the pairings II, IE, EI, and EE (first letter: clamped subdomain). Overlap decompositions split the bar lengthwise with overlap widths of 50.8, 25.4, and 12.7 mm; the nonoverlap decomposition splits it at  $x = 127$  mm. The free subdomain keeps  $h = 6.35$  mm and the clamped subdomain is coarsened to  $6.35/r$  mm; the families used are  $r = 1.0, 0.9, \dots, 0.2$  (the mesh-ratio sweep of 2026-07) and  $r \in \{0.9, 0.75, 0.5\}$  (the 2026-09 rerun with the current code and the Robin–Robin study). The Schwarz iteration is converged to a relative tolerance of  $10^{-8}$  at every stop.

**Nested cylinders.** A shell (outer radius 88.9 mm, height 330 mm, wall 12.7 mm;  $E = 200$  GPa,  $\rho = 7823$  kg/m<sup>3</sup>) contains a soft filler (radius 76.2 mm, height 305 mm; bulk modulus 5.99 GPa, shear modulus 71.7 MPa,  $\rho = 1843$  kg/m<sup>3</sup>) that contains an offset core (radius 38.1 mm, height 102 mm, 19 mm above center;  $E = 100$  GPa,  $\rho = 4915$  kg/m<sup>3</sup>). All three are neohookean. The P-wave impedance of the filler, 3.35 MPas/m, is one fiftieth of the shell's. The bottom face is fixed in  $z$  and the whole assembly is launched at  $-35$  m/s in  $z$ , so a compressive front runs up through the assembly. The decomposition splits the filler at the boundary of a cylindrical region around the core (radius 57 mm, height 229 mm): the interface has a lateral surface and two caps and lies in one material. The overlap version grows the inner region and shrinks the outer cavity by 7.6 mm each. The inner mesh is 5.1 mm and the outer 6.35 mm (ratio 1:0.8, nonconforming), 17 to 26 thousand elements per subdomain; the monolithic reference has 59 thousand elements at 5.1 mm. Every subdomain uses central difference with  $0.5 \mu$ s stops (Courant number 0.5) to a horizon of 5 ms. One Cubit journal with a refinement parameter generates all meshes (`examples/jmp/concentric-cylinders`).

**Linear nested cylinders.** For devising and testing conservation schemes the cylinders also exist with linear elastic materials (`examples/jmp/concentric-cylinders-linear`, the same meshes by relative path): core  $E = 100$  GPa,  $\nu = 0.33$ ; shell  $E = 200$  GPa,  $\nu = 0.28963$ ; filler  $E = 9KG/(3K + G) = 0.21412$  GPa and  $\nu = (3K - 2G)/(2(3K + G)) = 0.49404$  from its bulk and shear moduli; the same densities. With linear materials  $E_2$  is defined. The assembly moves at  $-5$  m/s rather than  $-35$  m/s, which strains the filler beyond the range of a small-strain model; the conservation identity does not depend on the amplitude, and the runs below repeat the case at  $-35$  m/s. Central difference at Courant number 0.5,  $0.5 \mu$ s stops, 1 ms. The nonoverlap input uses the constrained exchange with the inner (finer) side as the Dirichlet side.

**Explicit stability.** Central difference in Norma caps its step at  $\text{CFL} \times h_{\min}/c_p$ , where  $h_{\min}$  is the smallest element characteristic diameter (twice the mean distance from the element centroid to its nodes) over the block and  $c_p$  the P-wave speed of the block material; a stop longer than the cap is reached by substeps. On the cantilever meshes that diameter is about 1.7 times the edge length, so the cap at  $\text{CFL} = 1$  exceeds the true stability limit; the cantilever explicit runs at  $1 \mu$ s operate at 0.27 of the measured limit ( $3.6 \mu$ s on the 6.35 mm mesh) and are unaffected by it. A Courant number of 0.5 or less makes the cap protective on hexahedral meshes, and the cylinders use 0.5.

## 1.4 Energy metrics

**Blended energy.** Summing the energies of the two subdomains counts the overlap region twice, and the double count alone produces apparent excursions of  $\pm 13\%$  as a pulse crosses the overlap. All energies in this note count the overlap once through a partition of unity: the production output (`blended energy output: true, src/overlap-energy.jl`) weights each subdomain by the ratio of its distance to its own Schwarz boundary to the sum of both distances, a smooth Arlequin partition, and nonoverlap decompositions receive unit weights. Results are reported as  $E(t)/E(0)$  or as the drift  $E(t)/E(0) - 1$  in percent.

**Staggered energy for central difference.** Central difference conserves, on linear problems, the staggered lumped-mass energy

$$E_n = \frac{1}{2} \dot{\mathbf{u}}_{n-\frac{1}{2}}^T \mathbf{M}_L \dot{\mathbf{u}}_{n+\frac{1}{2}} + \text{SE}(\mathbf{u}_n) = \frac{1}{2} \dot{\mathbf{u}}_n^T \mathbf{M}_L \dot{\mathbf{u}}_n - \frac{\Delta t^2}{8} \mathbf{a}_n^T \mathbf{M}_L \mathbf{a}_n + \text{SE}(\mathbf{u}_n), \quad (3)$$

with  $\mathbf{M}_L$  the lumped mass and SE the stored energy. The consistent-mass kinetic energy at integer steps misrepresents the short-wavelength content of a sharp front: a monodomain central-difference run of the cantilever reads  $-8.2\%$  at 1 ms in that metric and  $\pm 0.000\%$  in (3). Norma's energy output evaluates (3) for central-difference subdomains, and every explicit result below uses it. The correction term is negative, so when grid-frequency content accumulates at the stability limit ( $\omega\Delta t \rightarrow 2$ ) the metric goes negative while displacements and stored energy stay bounded; the note calls that state *bounded and noisy*.

## 2 Transmission conditions on a shared interface

### 2.1 Robin coupling

On each side of the interface a Robin parameter  $\alpha > 0$  is prescribed and the condition

$$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad k = 1, 2, \quad (4)$$

is imposed, with the datum built from the partner's previous Schwarz iterate,  $\mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}$ . At the fixed point  $\mathbf{u}_k = \mathbf{u}_j$  and  $\mathbf{t}_k = -\mathbf{t}_j$ , so (4) reproduces (1) and (2). Discretely, the self term adds the boundary stiffness  $\alpha \mathbf{W}$  to the tangent and the partner datum enters the boundary force as  $-\mathbf{t}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$ . The parameter  $\alpha$  has units of stiffness per unit area ( $\text{N/m}^3$ ) and interpolates between a Dirichlet constraint ( $\alpha \rightarrow \infty$ ) and a Neumann condition ( $\alpha \rightarrow 0$ ); the classical choice is  $\alpha \sim E/h$ .

The input keyword is `Schwarz RR nonoverlap`. It requires a positive `robin parameter`, rejects `impedance scale`, defaults to `adjoint pairing: false` (the classical per-side transfer, which allows different  $\alpha$  on the two sides), and warns when combined with a dynamic integrator, because the condition reflects every wave (Sec. 2.3) and pumps energy at a dynamic interface (Secs. 6.7 and 7). It is intended for quasi-statics and for comparison studies. Both `nonoverlap` keywords are implemented by the same boundary-condition type; the Robin condition is the  $Z = 0$  member of the impedance family, and the two keywords differ only in defaults and input guards.

### 2.2 Impedance coupling

For a semi-infinite elastic bar with density  $\rho$  and P-wave modulus  $M = \lambda + 2\mu$ , an outgoing wave  $u(x, t) = f(x - c_p t)$  with  $c_p = \sqrt{M/\rho}$  satisfies

$$\sigma + Z \dot{u} = 0, \quad Z = \rho c_p = \sqrt{\rho(\lambda + 2\mu)}, \quad (5)$$

the Lysmer–Kuhlemeyer nonreflecting condition [3]. The impedance Schwarz condition adds this absorbing term to (4):

$$\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad \mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + Z \dot{\mathbf{u}}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}. \quad (6)$$

At the fixed point the velocity and displacement terms cancel and (6) again reproduces continuity and traction balance. The impedance is computed from the material of the element block that owns the interface side set,  $Z_k = \sqrt{\rho_k(\lambda_k + 2\mu_k)}$ , multiplied by the input `impedance scale` (default 1, required positive);  $\alpha$  is an optional input (default 0). The keyword is `Schwarz impedance nonoverlap`, with `adjoint pairing: true` as its default (Sec. 5).

Under Newmark integration the end-of-step velocity is linear in the end-of-step displacement with coefficient  $c_v = \gamma/(\beta \Delta t)$ , so the self term contributes the tangent  $(Z c_v + \alpha) \mathbf{W}$  and the

self force  $\mathbf{W}(Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$ . In the quasi-static limit  $c_v = 0$  and the condition reduces to (4); a quasi-static run with  $\alpha = 0$  is a pure Neumann condition and does not couple the subdomains kinematically.

### 2.3 Reflection at the interface

Place the interface at  $x = 0$  with  $\Omega_k$  in  $x < 0$  and consider a harmonic incident wave and its reflection. The Robin condition  $\sigma + \alpha u = 0$  gives the reflection coefficient

$$R_{\text{RR}}(\omega) = \frac{i\omega Z - \alpha}{i\omega Z + \alpha}, \quad |R_{\text{RR}}(\omega)| \equiv 1, \quad (7)$$

so for any real  $\alpha > 0$  the interface reflects the incoming wave completely, with a frequency-dependent phase shift. The impedance condition  $\sigma + Z\dot{u} + \alpha u = 0$  gives

$$R_{\text{imp}}(\omega) = -\frac{\alpha}{2i\omega Z + \alpha}, \quad |R_{\text{imp}}(\omega)| = \frac{\alpha}{\sqrt{\alpha^2 + 4\omega^2 Z^2}} < 1, \quad (8)$$

with  $R_{\text{imp}} \equiv 0$  at  $\alpha = 0$ . The dashpot is the only mechanism in either condition that removes energy from an interface mode; the Robin spring stores it and returns it.

## 3 Transmission conditions on an overlap boundary

### 3.1 Dirichlet coupling

The classical overlap condition imposes the partner's kinematic trace on  $\Gamma_k$ ,

$$\mathbf{u}_k = \mathbf{u}_j, \quad \dot{\mathbf{u}}_k = \dot{\mathbf{u}}_j, \quad \ddot{\mathbf{u}}_k = \ddot{\mathbf{u}}_j \quad \text{on } \Gamma_k, \quad (9)$$

either by pointwise interpolation (locate the partner element that contains each node in the reference configuration and evaluate the partner shape functions there) or by the  $L^2$  projection  $\mathbf{P}_{kj} = \mathbf{W}_k^{-1} \mathbf{L}_{kj}$  with  $\mathbf{L}_{kj} = \int_{\Gamma_k} \mathbf{N}_k (\mathbf{N}_j^{\text{vol}})^\top d\Gamma$ . Both are implemented by `SolidMechanicsOverlapSchwarzBoundaryCondition` (keyword `Schwarz overlap`). The constrained degrees of freedom add no interface tangent.

### 3.2 Impedance coupling

The impedance overlap condition leaves the overlap degrees of freedom free and imposes

$$\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z}(\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha(\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0} \quad \text{on } \Gamma_k, \quad (10)$$

where  $\mathbf{t}_j = \boldsymbol{\sigma}_j \mathbf{n}_k$  is the partner's traction on  $\Gamma_k$  with the outward normal of  $\Omega_k$  (the partner has no boundary there, so the converged state satisfies  $\mathbf{t}_k = \mathbf{t}_j$ ). The own traction  $\mathbf{t}_k$  is never computed: it is the natural boundary term of the weak form of  $\Omega_k$ , and the condition is imposed by adding the partner datum  $\mathbf{W}(\mathbf{t}_j + \mathbf{Z} \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$  as a Neumann load and the self terms  $\mathbf{W}(Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$  as an interface force on the same residual. The impedance is the P/S-split tensor

$$\mathbf{Z} = Z_p \mathbf{n}_k \otimes \mathbf{n}_k + Z_s (\mathbf{I} - \mathbf{n}_k \otimes \mathbf{n}_k), \quad Z_p = \sqrt{\rho(\lambda + 2\mu)}, \quad Z_s = \sqrt{\rho\mu}, \quad (11)$$

with  $\rho$ ,  $\lambda$ ,  $\mu$  from the partner's material in the block named by `source block` (each side carries the impedance of its neighbor, the optimized-Schwarz principle that the transmission operator should approximate the neighbor's Dirichlet-to-Neumann map). The keyword is `Schwarz impedance overlap`.

**The partner traction.** The monodomain solution has continuous kinematics and a nonzero traction on the artificial surface  $\Gamma_k$ , and satisfies (10) identically. Without the term  $-\mathbf{t}_j$  the condition becomes a relative dashpot that the monodomain solution fails by exactly that traction, and the converged coupled solution then carries a spurious velocity jump of about  $\mathbf{t}/Z$  whose dissipated power  $|\mathbf{t}|^2/Z$  acts as permanent interface damping (59% loss over 1 ms on the conforming cantilever; Sec. 6.1). Because  $\Gamma_k$  is an interior surface of the partner mesh, the partner’s assembled internal force cannot supply  $\mathbf{t}_j$  there. Norma evaluates it in one of two ways (**partner traction**, default **auto**). On node-aligned interfaces the weak traction is the consistent nodal reaction of the single layer of partner elements exterior to  $\Gamma_k$ ,

$$[\mathbf{W} \mathbf{t}_j]_i = - \sum_{e \in \mathcal{E}_i^{\text{ext}}} (\mathbf{f}_e^{\text{int}} + \mathbf{M}_e \ddot{\mathbf{u}}_e)_i, \quad (12)$$

which the monodomain solution satisfies identically, mesh-scale content included (implemented by `build_consistent_traction_patch!`). Otherwise the partner’s consistently recovered nodal stress is transferred to  $\Gamma_k$  and contracted with the outward normal; recovery is enabled automatically for models that carry this condition.

**Transfer and assembly.** The kinematic fields and the recovered stress are transferred by the  $L^2$  projection by default (**transfer: variational**; pointwise interpolation is a legacy option, and the two coincide on node-aligned interfaces). On non-aligned interfaces the whole partner datum is assembled on the partner’s offset surface from the partner’s own nodal values and transferred with one surface-to-surface operator (**characteristic\_partner\_terms**). The self tangent consists of the nodal blocks  $W_{ij} (c_v \mathbf{Z}(\mathbf{n}_j) + \alpha \mathbf{I})$ . A scalar or step-dependent multiplier on  $\mathbf{Z}$  (**impedance scale**) remains available; with the traction term in place the physical scale 1 is the conserving one, and the historical use of the scale as a per-problem parameter, which compensated the missing traction, is obsolete. With **impedance scale: 0** the condition is the Robin–Robin overlap condition  $\mathbf{t}_k - \mathbf{t}_j + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0}$ ; no separate keyword exists for it.

### 3.3 Summary of the four conditions

Table 1: Transmission conditions on a shared (nonoverlap) interface.

|                    | Robin                                                       | Impedance                                                                                                |
|--------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Boundary condition | $\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$         | $\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$                               |
| Self tangent       | $\alpha \mathbf{W}$                                         | $(Z c_v + \alpha) \mathbf{W}$                                                                            |
| Self force         | $\alpha \mathbf{W} \mathbf{u}_k$                            | $\mathbf{W} (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$                                                |
| Partner datum      | $-\mathbf{t}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$ | $-\mathbf{t}_j + Z \mathbf{W} \mathbf{P} \dot{\mathbf{u}}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$ |
| $Z$                | (none)                                                      | interface block material; harmonic mean per pair                                                         |
| $\alpha$           | input, required, per side                                   | input, optional, one value per interface                                                                 |
| Transfer           | per side (legacy)                                           | adjoint pair from one cross-mass matrix                                                                  |
| Reflection $ R $   | $\equiv 1$                                                  | $< 1$ ; 0 at $\alpha = 0$                                                                                |
| Keyword            | Schwarz RR nonoverlap                                       | Schwarz impedance nonoverlap                                                                             |

Table 2: Transmission conditions on an overlap boundary.

|                    | Dirichlet                                           | Impedance                                                                                                                              |
|--------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Boundary condition | $\mathbf{u}_k = \mathbf{u}_j$ (pointwise or $L^2$ ) | $\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z}(\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha(\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0}$ |
| Self tangent       | (DOFs constrained)                                  | $W_{ij}(c_v \mathbf{Z}(\mathbf{n}_j) + \alpha \mathbf{I})$                                                                             |
| Self force         | (none)                                              | $\mathbf{W}(\mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$                                                                      |
| Partner datum      | $\mathbf{u}_k \leftarrow \mathbf{P} \mathbf{u}_j$   | $\mathbf{W}(\mathbf{t}_j + \mathbf{Z} \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$                                                       |
| Partner traction   | (none)                                              | consistent (12) or recovered stress                                                                                                    |
| $\mathbf{Z}$       | (none)                                              | partner material, P/S split                                                                                                            |
| $\alpha$           | (none)                                              | input, optional                                                                                                                        |
| Keyword            | Schwarz overlap                                     | Schwarz impedance overlap                                                                                                              |

## 4 Requirements for energy consistency

Three results from the literature fix the structure of an energy-consistent coupling; the measurements of Secs. 6–7 confirm each of them.

**The interface work identity.** For Newmark-family subdomains coupled by interface tractions, the discrete energy balance over a step contains the interface term (Gravouil–Combescure [4, Eq. (34)]; Karimi–Nakshatrala [13, Eq. (84)])

$$E_{\text{int}} = \sum_{\text{sides } k} \frac{1}{\Delta t_k} \llbracket \dot{\mathbf{u}}_k \rrbracket^T \mathbf{C}_k^T \llbracket \Lambda_k \rrbracket, \quad (13)$$

the work of the transmitted tractions through the kinematic jump, each side on its own grid ( $\mathbf{C}_k$  is the interface restriction of side  $k$ ,  $\Lambda_k$  the traction impulse transmitted to it, and  $\llbracket \cdot \rrbracket$  here the change across the step). The term vanishes if and only if the traction is one field acting equal and opposite on the two sides through adjoint trace operators, with velocity continuity enforced at instants shared by both time grids. When the pairing is broken the term has no sign.

**Representability on nonconforming grids.** Gander, Halpern, and Nataf [14] prove convergence of Schwarz waveform relaxation with impedance transmission on nonmatching grids under three conditions: Robin-type (not Dirichlet) transmission, a consistent boundary discretization of the transmission operator, and an intergrid transfer that is an  $L^2$  contraction (a projection qualifies; pointwise interpolation does not). The choice  $\mathbf{Z} = \rho \mathbf{c}$  is the exact transparent operator in one dimension. On nonconforming grids the coupled system defines the converged solution; there is no monodomain scheme it equals. Continuity holds only through the projection, and the component of the fine trace orthogonal to the coarse trace space is never matched: the dashpot then absorbs that persistent component as an absorbing boundary would, which is sign-definite dissipation, required for stability. Achdou, Japhet, Maday, and Nataf [15] locate the remedy in a common trace space for the datum, not in a better  $\mathbf{Z}$ .

**The adjoint requirement.** Energy-stable discontinuous Galerkin couplings [16, 17, 18] compute one interface state on a common description of the interface and use it in both weak forms; the interface power then reduces to

$$P = - \int_{\Gamma} \llbracket \dot{\mathbf{u}} \rrbracket^T \mathbf{Z} \llbracket \dot{\mathbf{u}} \rrbracket dS \leq 0, \quad (14)$$

which is confined to the jump and vanishes under refinement and at convergence. The discrete requirement is that the two kinematic transfers be  $L^2$ -adjoint through one cross-mass matrix,

$$\mathbf{W}_1 \Pi_{2 \rightarrow 1} = (\mathbf{W}_2 \Pi_{1 \rightarrow 2})^T = \mathbf{B}, \quad B_{mn} = \int_{\Gamma} \varphi_m^1 \varphi_n^2 dS. \quad (15)$$

Two independent interpolations violate (15), the cross terms of the interface power do not cancel, and its sign is undetermined. Liu et al. [18] add that interface unknowns may need implicit treatment even between explicit interiors.

**Consequence for the two decompositions.** Condition (15) presumes one shared surface on which the two works cancel. The overlap decomposition has none: data are exchanged on two distinct surfaces  $\Gamma_1$  and  $\Gamma_2$ , so no pairing of the two works exists, and energy consistency could only come from agreement of the two discrete solutions in the overlap, which the differing discrete dispersion relations of the two grids preclude at fixed  $h$  (Sec. 6.3). The nonoverlap decomposition has a shared interface, and the adjoint-paired coupling of Sec. 5 implements (15) on it.

## 5 The adjoint-paired nonoverlap impedance coupling

This section describes the coupling as implemented (the default of `Schwarz impedance nonoverlap; adjoint pairing: false` restores the legacy per-side transfer). Its measured behavior is in Secs. 6.4 and 7.

### 5.1 Construction

**Transfer operators.** The cross-mass matrix  $\mathbf{B}$  of (15) is integrated over the facets of the finer side (the side with more interface nodes): each facet quadrature point is mapped to the partner surface by closest-point projection in the reference configuration and the partner facet shape functions are evaluated there. This rule is exact when the coarser trace mesh is nested in the finer facets and carries facet-quadrature error where a coarse basis function kinks inside a fine facet; the partition-of-unity error  $\max_i |(\Pi_k \mathbf{1} - \mathbf{1})_i|$  is logged at setup as a certificate (zero on conforming and nested interfaces). The kinematic transfers are  $\Pi_1 = \mathbf{W}_1^{-1} \mathbf{B}$  and  $\Pi_2 = \mathbf{W}_2^{-1} \mathbf{B}^T$ , the force transfers their partner adjoints  $N_1 = \Pi_2^T$  and  $N_2 = \Pi_1^T$ , all scalar and applied to each Cartesian component.

**Exchanged traction.** The weak partner reaction of side  $j$  is the restriction to its interface degrees of freedom of its assembled d'Alembert residual,

$$\lambda_j = -(\mathbf{M}_j \ddot{\mathbf{u}}_j + \mathbf{f}_j^{\text{int}} - \mathbf{f}_j^{\text{body}})|_{\Gamma}, \quad (16)$$

with  $\mathbf{M}_j$  the mass matrix its integrator uses (consistent for Newmark, lumped for central difference). With this reaction the two interface rows sum exactly to the monodomain row, so the monodomain trajectory is the fixed point. The legacy coupling transferred the static reaction  $-\mathbf{f}^{\text{int}}$ ; in dynamics the two sides' static reactions differ by the inertia of the interface nodes, each side carrying its own tributary mass, and the fixed point then differs from the monodomain discretization by a steady, step-independent interface energy loss ( $-9.6\%$  per millisecond on the conforming cantilever, Table 4).

**Pair impedance and Robin parameter.** The pair shares one impedance, the harmonic mean  $2Z_1Z_2/(Z_1 + Z_2)$  of the per-side material impedances  $Z_k = \rho_k c_{p,k}$  of the blocks that own the two side sets, each multiplied by its side’s **impedance scale**; this is the Riemann-solver weighting of [17, 18]. The Robin parameter  $\alpha$  is one value per interface: the spring stores and returns energy conservatively only when both interface forces pair through the same coefficient, so mismatched values abort with a message naming both side sets. The pair impedance is scalar (P-wave); a tensor pair impedance has not been needed on the benchmarks studied.

**Partner datum.** Side  $k$  receives

$$\mathbf{g}_k = N_k \lambda_j + Z \mathbf{B}_k \dot{\mathbf{u}}_j + \alpha \mathbf{B}_k \mathbf{u}_j, \quad \mathbf{B}_1 = \mathbf{B}, \quad \mathbf{B}_2 = \mathbf{B}^T, \quad (17)$$

with the partner fields evaluated at side  $k$ ’s current substep time (Sec. 5.3). The self terms, the tangent  $(Zc_v + \alpha) \mathbf{W}_k$  and the force  $\mathbf{W}_k(Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$ , are assembled with the internal force at every residual evaluation.

**What the parameters change.** At the fixed point the two conditions add to traction balance and subtract to  $Z \dot{\boldsymbol{\delta}} + \alpha \boldsymbol{\delta} = \mathbf{0}$  for the interface jump  $\boldsymbol{\delta}$ . When the jump can close, as it does on conforming and nested meshes, the converged solution is independent of  $Z > 0$  and  $\alpha \geq 0$ , and both act only on the convergence rate of the iteration. On nonconforming meshes the jump closes only in the projected sense of (15) and the accepted iterate has a residual jump, so the  $\alpha$  term is a spring force on that residual, and its influence on the converged coupled solution has not been measured for the paired coupling (Sec. 9.3). The classical Robin–Robin runs of Sec. 6.7 show that with per-side transfer the converged solution does depend on  $\alpha$ .

## 5.2 Explicit subdomains

With a central-difference subdomain on either side, the coupling is strongly unstable if the interface terms are evaluated at lagged time levels: the exchanged reaction (16) couples the partner’s acceleration into the interface force, an algebraic loop that the implicit pairs resolve within the Newton iteration. Following Liu et al. [18], the explicit update is closed implicitly on the interface rows only,

$$(\mathbf{M} + \gamma \Delta t Z \mathbf{W})|_{\Gamma} \mathbf{a}_{n+1} = \mathbf{M} \mathbf{a}^{\text{expl}}|_{\Gamma}, \quad (18)$$

a small symmetric positive definite solve per interface, factored once and reused for the three components while the operator is unchanged (`imex_interface_acceleration!`); the interior stays explicit. Here  $\mathbf{M}|_{\Gamma}$  is the lumped mass on the interface rows,  $\mathbf{W}$  the side’s own boundary mass matrix,  $\gamma = \frac{1}{2}$ , and  $\mathbf{a}^{\text{expl}}$  the plain explicit update, whose right-hand side contains the dashpot at the predictor velocity and the Robin spring at the known end-of-step displacement; the solve replaces the predictor-velocity dashpot by the end-of-step one. The Robin spring stays explicit, so a large  $\alpha$  can violate the central-difference stability limit on the interface rows. Interface degrees of freedom held by Dirichlet conditions keep their prescribed acceleration and enter the free rows as data.

## 5.3 Different time steps per subdomain

Each subdomain records full-field snapshots (displacement, velocity, acceleration, assembled internal force) at every substep of every Schwarz iteration, anchored by a snapshot of the stop-start

state. When a coupling condition is evaluated at time  $t$ , each field is interpolated piecewise linearly to  $t$  (queries beyond the history clamp to its endpoints), the partner state is temporarily replaced by the interpolants, the coupling terms are evaluated, so the reaction (16) is formed from the interpolated acceleration and internal force, and the partner state is restored. This is the interpolated-continuity structure under which Gravouil and Combescure [4] prove the interface work of the subcycled exchange nonpositive and of order  $\Delta t$ . Two earlier defects (an off-by-one in the segment selection and a missing stop-start anchor) had reduced the exchange to end-sampled constant data; both are fixed and the measurements below use the interpolated exchange.

Three constructions that would make the coarse side’s datum exactly work-conjugate to the fine side’s interpolation, as in Prakash and Hjelmstad [6], were implemented and all fail in a partitioned exchange: the trapezoidal window average lags the interface motion by half a window and adds dissipation; the recursion  $\hat{\lambda}_{n+1} = 2\bar{\lambda} - \hat{\lambda}_n$  doubles the contraction factor of the Schwarz iteration and diverges within tens of stops; and the least-squares endpoint fit is an extrapolation whose gain exceeds one outside the linear band and injects energy without bound over hundreds of stops. Only causal transfer operators are robust in a fixed-point exchange of the impedance coupling. The constrained Dirichlet–Neumann exchange cancels the subcycled interface work without a direct solve, with the coarse-step side as the Dirichlet side (Sec. 8.2). The window-average and endpoint-fit utilities remain in `interpolation.jl` as tested building blocks.

## 5.4 Relaxation and Aitken acceleration

The relaxed datum is the assembled  $\mathbf{g}_k$  of (17) on one side of each pair (the later-listed subdomain); a fresh relaxation slot starts from zero rather than from the incoming datum. Two Aitken forms are available (`relaxation: aitken recursive` and `aitken secant`), with `relaxation parameter` the fixed or initial factor and `aitken NO parameter` the number of initial Schwarz iterations at the input factor.

**State keyed by substep slot.** The relaxation state (the stored datum and the Aitken residuals) is keyed per pair and per substep time slot within the stop, and cleared at every stop (`relaxation_slot!`). Before this, one vector per pair was updated at every application of the condition, so a stop spanning several substeps blended iterates across time, a causal low-pass filter on the exchanged traction that shifted the fixed point (measured on the conforming cantilever at 10 ms: +11% to −84% with Aitken and −7% to −88% with fixed  $\theta = 0.5$  as the window grew from 2 to 10 substeps). With the slot-keyed state, windowed stops reproduce the same-step trajectory to Schwarz truncation (Sec. 6.5).

**Guards.** On the Dirichlet-transfer paths (overlap and Dirichlet–Neumann) a fresh slot seeds the stored iterate with the incoming datum, so the first residual is exactly zero and the secant factor evaluates to −0.0; a zero factor freezes the interface, and in a chain of three or more subdomains the whole Schwarz iteration then reports convergence after one iteration with an answer wrong by 100%. The secant update is now taken only when the residual difference exceeds a floor, a factor of magnitude below  $10^{-12}$  is recorded on the controller, and the convergence criterion refuses to accept that iteration. The recursive form no longer runs the first NO iterations unrelaxed regardless of the input factor.

**Where Aitken applies.** Aitken acceleration is applied only to single-slot stops (`aitken_applies`); in windowed stops the residuals of different slots belong to different times and the extrapolated factor overshoots. On the implicit cantilever pairs it reduces the Schwarz iterations by about a

factor of ten against a fixed factor of 0.5 (599 against 56 recursive or 58 secant over ten stops). On the explicit cylinders pair (Sec. 7) every Aitken form diverges: the displacement residual between successive iterations collapses while the interface velocity jump stalls at a finite value, the extrapolated factor exceeds 2 by the second or third iteration, oscillates between  $-10$  and  $+3$ , and the iteration reaches the cap of 256 within the first 25 stops, after which an element inverts; the recursive form with  $N0 = 2$  and the secant form fail the same way, six stops later. A fixed factor of 0.5 converges there in 2 to 3 iterations per stop, and  $\theta = 1$  (no relaxation) diverges after 0.3 ms.

## 6 Measurements on the cantilever beam

### 6.1 Overlap, conforming meshes: the partner traction

On the conforming node-aligned overlap pair (II,  $\alpha = 0$ , physical impedance) the energy error of the impedance overlap coupling over 1 ms decreases through four corrections, none with a parameter:

$$\underbrace{59\%}_{\text{no partner traction}} \longrightarrow \underbrace{14\%}_{\text{traction from lumped recovery}} \longrightarrow \underbrace{1.5\%}_{\text{consistent recovery}} \longrightarrow \underbrace{0.08\%}_{\text{consistent traction (12)}}$$

against the 0.1% of converged conforming Dirichlet coupling. The residual of the recovered-stress step was diagnosed with an interface-work instrumentation that separates the partner-traction, dashpot, and Robin channels (`report_impedance_interface_work`, environment-gated, CSV through `NORMA_IMPEDANCE_WORK_CSV`): it is unchanged under a Schwarz tolerance of  $10^{-11}$  and under halving the time step, and it halves when  $h$  is halved. The recovery error on mesh-scale content is converted by the transmission condition into a velocity jump of about  $\delta t/Z$  that the dashpot dissipates at rate  $|\delta t|^2/Z$  over the whole run; the consistent traction removes it (integrated dashpot work  $-42.6 \text{ J} \rightarrow +0.1 \text{ J}$ ). The P/S tensor impedance (11), which appeared worse than a scalar one with recovered stress (4.1% against 1.5%), is exact with the consistent traction (0.076%): its smaller shear impedance amplified the tangential recovery error. Historical per-problem tuning of the impedance scale, whose best value on the nonconforming pair was 1.12, compensated the missing traction against transfer injection and is obsolete.

### 6.2 Overlap, nonconforming meshes: sign-indefinite interface work

Table 3 gives the drift at 1 ms of the impedance overlap coupling with variational transfer across nine mesh ratios, three overlap widths, and four integrator pairings (108 runs).

Conforming implicit runs conserve to 0.1% at every width. Every nonconforming configuration either loses tens of percent or gains: the II column injects at mild coarsening at nearly every width (up to  $+29.8\%$  at 1:0.8), the 25.4 mm IE run at 1:0.3 reaches  $+275\%$ , and the drift is not monotone in the ratio (at 25.4 mm the II column moves  $-52.7 \rightarrow +173.7 \rightarrow -66.5\%$  across 1:0.4, 1:0.3, 1:0.2). The sign of the interface work is decided by the mesh pair and the overlap geometry together, and no a priori bound on its sign or magnitude exists. The channel instrumentation shows the mechanism: on the 50.8 mm/1:0.5 II case with pointwise transfer the dashpot does  $+554 \text{ J}$  of positive work over 1 ms, which a dissipator cannot do, so the discrete dashpot term is not a quadratic form; the two sides interpolate partner fields with unrelated operators and their dashpot terms do not pair.

**Dirichlet control.** The same nine mesh configurations with Dirichlet overlap coupling (II) conserve to  $\pm 0.05\%$  on conforming meshes and are severely unstable on every nonconforming one ( $+22\%$  to  $+7765\%$  at 1 ms, most runs ending in Newton divergence before 1 ms), with pointwise and with weak transfer alike. The dashpot of the impedance condition is therefore what stabilizes

Table 3: Energy drift at 1 ms of the impedance overlap coupling (variational transfer, blended energy), in percent.

| ratio | overlap 50.8 mm |       |       |       | overlap 25.4 mm |                |       |       | overlap 12.7 mm |       |       |       |
|-------|-----------------|-------|-------|-------|-----------------|----------------|-------|-------|-----------------|-------|-------|-------|
|       | II              | IE    | EI    | EE    | II              | IE             | EI    | EE    | II              | IE    | EI    | EE    |
| 1:1.0 | -0.1            | -8.9  | -0.2  | +8.7  | -0.1            | +5.0           | -0.4  | +10.2 | -0.0            | -15.7 | -6.1  | +3.6  |
| 1:0.9 | +7.6            | -11.7 | -5.5  | +8.5  | -4.4            | -8.1           | +4.4  | +9.1  | +0.1            | -14.8 | -1.0  | +4.3  |
| 1:0.8 | + <b>29.8</b>   | -15.3 | +5.5  | -9.5  | +20.4           | + <b>45.1</b>  | -2.4  | -1.9  | -3.1            | -20.3 | -5.4  | +4.7  |
| 1:0.7 | +19.2           | -12.8 | +2.7  | -11.6 | +11.8           | -18.6          | +0.7  | -5.4  | +11.1           | +7.3  | -9.8  | -17.9 |
| 1:0.6 | -61.1           | -54.0 | -57.7 | -40.4 | +2.1            | -18.6          | -63.9 | -53.4 | -4.5            | -34.2 | -57.1 | -52.7 |
| 1:0.5 | -50.6           | -32.3 | -36.4 | -56.8 | -11.0           | -17.6          | -55.7 | -54.3 | -29.2           | -54.0 | -56.8 | -52.0 |
| 1:0.4 | -12.4           | +3.9  | -43.1 | -53.1 | -52.7           | -50.2          | -57.0 | -63.8 | -36.6           | -52.4 | -78.2 | -76.2 |
| 1:0.3 | -65.0           | -44.5 | -78.3 | -57.4 | + <b>173.7</b>  | + <b>274.7</b> | -81.2 | -78.8 | -30.5           | -58.6 | -79.5 | -79.4 |
| 1:0.2 | -79.1           | -76.8 | -80.8 | -78.9 | -66.5           | -66.2          | -86.3 | -87.4 | -22.2           | -45.8 | -91.2 | -90.3 |

a coupling that kinematic matching destabilizes, and its dissipation is the energy of the inter-grid discrepancy it absorbs.

**Ten milliseconds.** Of the 108 overlap runs rerun to 10 ms, 74 ended by element inversion after exponential energy growth (typically 30 to 100 times  $E(0)$  at failure, between 2 and 9 ms) and most of the remainder decayed toward zero energy. Configurations that appeared dissipative at 1 ms reversed and grew. Conformity does not stabilize the mixed pairings: the conforming 25.4 mm IE run grew exponentially and failed at 6.25 ms. The only overlap configurations stable and accurate at 10 ms are the conforming II runs (+0.6, +0.6, +0.2% at the three widths; Fig. 1).

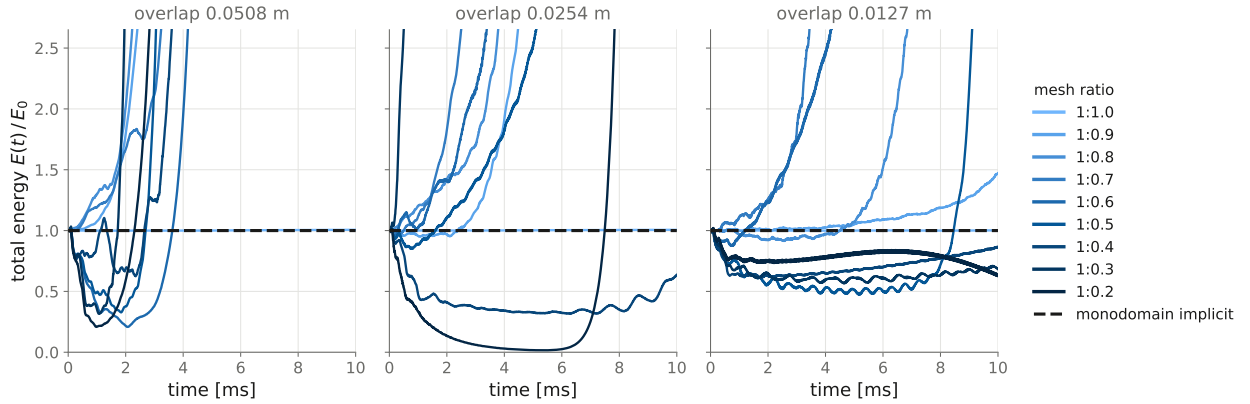


Figure 1: Overlap impedance coupling (variational transfer), implicit–implicit, 10 ms. Nearly every nonconforming configuration grows exponentially until element inversion (curves leaving the top of the axes) or decays toward zero energy; only the conforming runs follow the monodomain reference.

### 6.3 Overlap: remedies tried on the interface

Every remedy applied to the overlap exchange was measured on the instrumented 1 ms benchmark; none is a structural remedy, and the sequence explains why.

- **Transfer fidelity.** Replacing pointwise interpolation by the variational projection, refining

the transfer quadrature, taking the consistent traction across the interface from the partner’s offset surface, and transferring the whole partner datum with one operator each *increased* the spurious dashpot work on the 1:0.75 pair (−607, −726, −857, −1129, −1084 J) while the net drift moved from −0.75% to +9.8%: pointwise interpolation and nodal stress recovery had been smoothing the transferred data, and the small net of the original scheme was a cancellation of two large channels. Over the 24 nonconforming cases of the three-ratio family the variational transfer improved 17 and reduced the mean absolute drift from 33 to 25%; it is the default because it is the contraction the convergence theory requires, not because it conserves.

- **Content-aware absorption** (`content aware absorption`): an additional dissipative term  $\mathbf{W} \mathbf{Z} (\mathbf{I} - \hat{\Pi}) \dot{\mathbf{u}}_k$ , with  $\hat{\Pi}$  the  $\mathbf{W}$ -orthogonal projection onto the transferable subspace, is verified (annihilates constants, vanishes on node-aligned interfaces, provably dissipative; test 106) and has no effect: it dissipates 4.6 J while every channel stays identical. The recirculating content lies inside the transferable subspace: both grids represent it, and the discrepancy comes from their different discrete dispersion relations, which no transfer construction can remove.
- **Representable dashpot** (`representable dashpot`): restricting the dashpot to  $\mathbf{Z} \hat{\Pi} [\dot{\mathbf{u}}]$  leaves the dashpot work unchanged where the projection is nontrivial (−1066 → −1065 J) and worsens the drift where it acts (+12.5 → +45.6%). Retained as a diagnostic, default off.
- **Numerical damping**. Newmark  $\gamma = 0.6$  suppresses the recirculation but removes 87% of the physical energy. HHT- $\alpha$  (`HHT alpha`,  $\bar{\alpha} \in [0, 1/3]$ ) at  $\bar{\alpha} = 0.02$  to 0.05 converts the sign-indefinite drift into monotone decay with a loss proportional to the energy in the grid-frequency band; on this benchmark, whose release has content at every scale, the loss is 5 to 11% at 1 ms.

## 6.4 Nonoverlap: legacy and paired coupling

Table 4: Energy drift at 1 ms of the nonoverlap impedance coupling on the cantilever, in percent. Left: legacy per-side transfer against the adjoint-paired coupling, implicit–implicit. Right: the paired coupling in all four integrator pairings, staggered metric for explicit members.

| ratio | legacy II | paired II     | paired II | IE    | EI    | EE    |
|-------|-----------|---------------|-----------|-------|-------|-------|
| 1:1.0 | −9.61     | − <b>0.03</b> | −0.03     | −0.11 | +0.20 | −0.27 |
| 1:0.9 | −6.84     | − <b>0.03</b> | −0.03     | −0.10 | +0.29 | −0.27 |
| 1:0.8 | −7.96     | −3.59         | −3.6      | −4.4  | −3.5  | −7.3  |
| 1:0.7 | −2.06     | −6.55         | −6.6      | −6.2  | −5.3  | −9.8  |
| 1:0.6 | −9.39     | −9.71         | −9.7      | −9.8  | −9.8  | −11.8 |
| 1:0.5 | −13.25    | −13.32        | −13.3     | −13.8 | −10.2 | −10.5 |
| 1:0.4 | −45.78    | −15.13        | −15.1     | −17.8 | −11.4 | −13.3 |
| 1:0.3 | +18.95    | −15.08        | −15.1     | −11.5 | −26.4 | −20.2 |
| 1:0.2 | +77.49    | −13.26        | −13.3     | −20.1 | −27.8 | −21.4 |

The legacy coupling loses 2 to 18% at mild contrast in every pairing, including on conforming meshes (−9.6% II at 1:1, the static-reaction defect of Sec. 5.1, unchanged under  $\alpha \rightarrow 0$ , equal  $\alpha$ , and  $\Delta t/2$ ), and injects strongly at strong contrast. The paired coupling (Table 4, Fig. 2) never injects at any ratio in any pairing, conserves to 0.3% on conforming and 1:0.9 meshes in all four

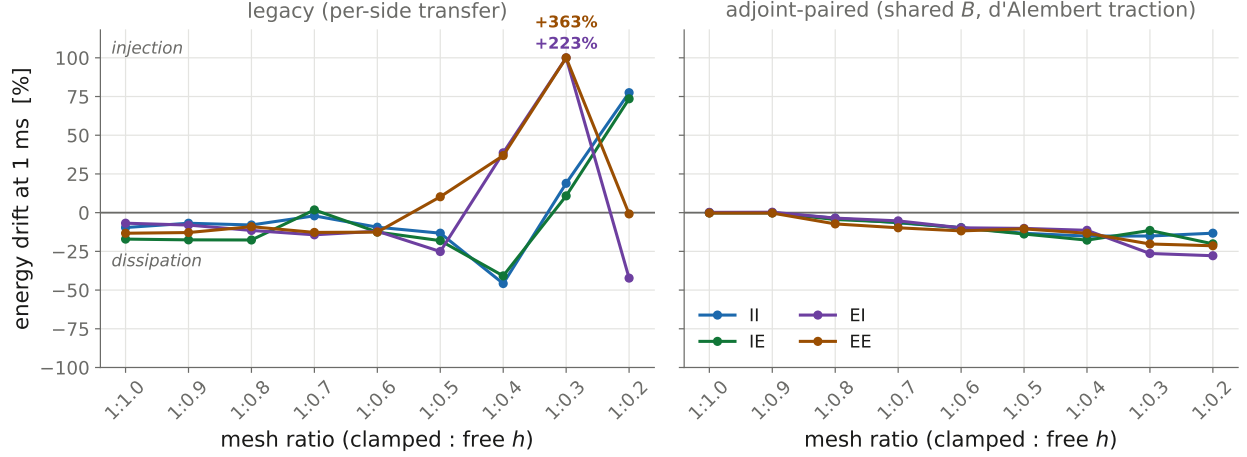


Figure 2: Nonoverlap impedance coupling, drift at 1 ms across the mesh-ratio family and the four integrator pairings. Left: legacy per-side transfer (consistent-mass metric), with injection up to +363% (EE) and +223% (EI) at 1:0.3. Right: adjoint-paired coupling (staggered metric for explicit members), sign-definite dissipation at every ratio and conservation to 0.3% on conforming and 1:0.9 meshes.

pairings, and against the monodomain explicit reference the conforming EE trajectory agrees to 0.35% pointwise over the millisecond. The residual dissipation at strong contrast is the interface jump the two trace spaces cannot represent in common, absorbed at rate  $Z \int_{\Gamma} [\dot{\mathbf{u}}]^2$ : at the nested ratio 1:0.5, where the legacy transfer is coincidentally adjoint and the quadrature exact, paired and legacy agree to 0.1%. The small positive EI entries are bounded transient storage in the interface spring.

**Ten milliseconds.** Table 5 and Fig. 3 give the 10 ms behavior. With an implicit member the conforming and 1:0.9 columns stay within 0.0 to  $-0.9\%$  over ten thousand stops. At strong contrast the dissipation that read 10 to 28% at 1 ms reaches 40 to 85%: bounded and monotone, but far too large for wave-transit accuracy, since absorption of unrepresentable content proceeds at a persistent rate. The non-nested mild ratios 1:0.7 and 1:0.8 form a failure window not detectable at 1 ms: interface-local high-frequency displacement content accumulates until an interface element inverts (implicit pairs, about 5 ms, with the global energy still smooth) or central difference becomes unstable (explicit members, 2.6 to 4 ms, at 0.27 of the stability limit, so driven by the accumulated content). Reruns with fixed  $\theta = 0.5$  fail identically, so the accumulation belongs to the transfer operators, whose partition-of-unity certificate is worst in this window; the nested and the strong ratios do not show it.

**Explicit–explicit pairs.** The EE column ends bounded and noisy at every ratio, conforming included: the stored energy stays physical (the conforming run keeps  $E/E(0) \geq 0.97$  to 5 ms) while the staggered kinetic term goes negative, because central difference with  $\gamma = \frac{1}{2}$  removes no grid-frequency content at any stable step and the exchange keeps introducing it. Subcycling the explicit member (free side at  $\Delta t/4$ ) removes the accumulation: that run ends at  $-10.8\%$ , the dissipation of the interpolated exchange. Subcycling has the opposite effect on implicit pairs, where the same-step conforming II run conserves to 0.00% and the subcycled one loses  $-20.2\%$  over  $10^4$  stops ( $-3.6\%$  at 1 ms), the interpolation dissipation of Sec. 5.3 accumulated over the run.

Table 5: Energy drift at 10 ms of the adjoint-paired nonoverlap coupling on the cantilever, staggered metric, in percent.  $\dagger t$ : element inversion at  $t$  ms. \*: bounded and noisy (Sec. 1.4).

| ratio | II            | IE            | EI            | EE            |
|-------|---------------|---------------|---------------|---------------|
| 1:1.0 | +0.00         | −0.9          | −0.2          | *             |
| 1:0.9 | +0.00         | −0.5          | +0.1          | *             |
| 1:0.8 | $\dagger$ 5.2 | $\dagger$ 3.9 | $\dagger$ 4.0 | $\dagger$ 2.6 |
| 1:0.7 | $\dagger$ 4.9 | $\dagger$ 3.7 | $\dagger$ 3.9 | $\dagger$ 2.8 |
| 1:0.6 | −47.3         | −56.4         | −42.9         | *             |
| 1:0.5 | −56.8         | −72.0         | −40.8         | *             |
| 1:0.4 | −54.9         | −73.9         | −40.4         | *             |
| 1:0.3 | −65.1         | −66.0         | −81.1         | *             |
| 1:0.2 | −53.3         | −73.1         | −84.1         | *             |

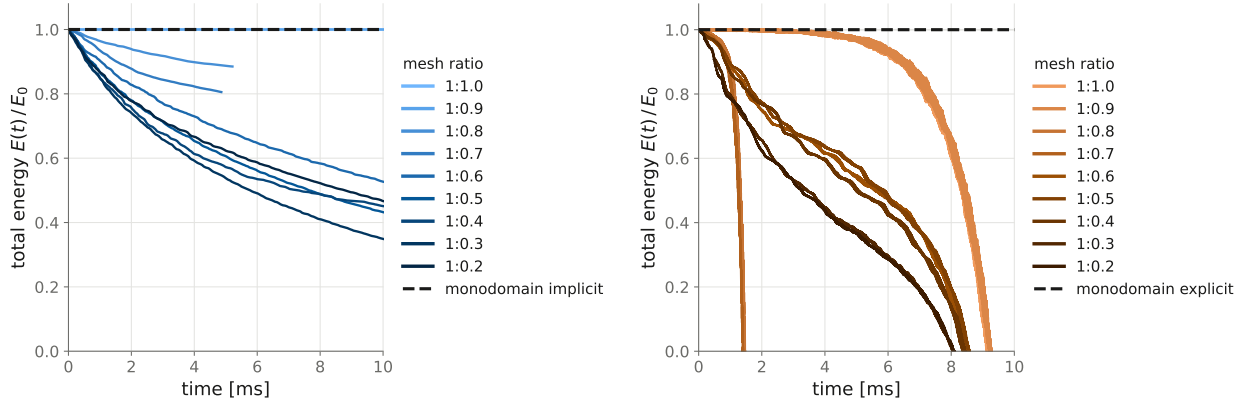


Figure 3: Adjoint-paired nonoverlap coupling, 10 ms, staggered metric. Left: implicit–implicit; no injection anywhere, conforming and 1:0.9 follow the monodomain reference, strong contrast dissipates monotonically, and 1:0.7 to 1:0.8 end at about 5 ms by element inversion. Right: explicit–explicit; the staggered kinetic term goes negative at every ratio as undamped grid-frequency content accumulates, and 1:0.7 to 1:0.8 fail by central-difference instability at 2.6 to 2.8 ms.

## 6.5 Windowed controller stops

With the subdomain steps fixed at  $1\mu\text{s}$ , controller stops of 1, 2, 5, and  $10\mu\text{s}$  were compared to 10 ms. For the overlap family every trajectory is identical across the four windows (its exchange has no relaxation state). For the nonoverlap coupling the pre-fix per-pair relaxation state shifted the fixed point (Sec. 5.4); with the slot-keyed state every windowed policy reproduces the same-step trajectory to Schwarz truncation (a displacement mismatch of  $3.9 \times 10^{-6}$  at working tolerances,  $3.7 \times 10^{-8}$  at tightened ones; regression in test 110). Inside a window the policies were measured on the conforming II case at the  $10\mu\text{s}$  window, in mean Schwarz iterations per stop:  $\theta = 1$  needs 19.1 and completes; per-slot Aitken about 25 and fails at 3.9 ms; fixed  $\theta = 0.5$  47.5; Aitken pooled over the window’s slots 55.1 on II and fails on EE. The total number of substep solves is proportional to the iterations per stop whatever the window, and all windowed counts exceed the same-step Aitken count, so the controller step is a convenience for the nonoverlap coupling (for aligning output stops) and never reduces work; for the overlap family larger stops reduce it slightly. The explicit–explicit pair at large windows is impractical under any policy (the windowed iteration is nearly non-contractive and reaches the iteration cap).

## 6.6 The five couplings with the current code

The five couplings (overlap Dirichlet, nonoverlap Dirichlet–Neumann, overlap impedance, paired nonoverlap impedance, and the monolithic reference) were rerun to 10 ms with the code of 2026-09-17 at ratios 1:0.9, 1:0.75, and 1:0.5 with overlap 25.4 mm (Table 6, Fig. 4). The Dirichlet–Neumann exchange, not part of the earlier study, ends by element inversion within 3 ms with implicit subdomains at every ratio and conserves within 1% to 10 ms with explicit ones; why the same exchange is unstable with implicit subdomains and stable with explicit ones has not been explained (Sec. 9.3). Every other entry reproduces the corresponding earlier result.

Table 6: Cantilever, overlap 25.4 mm, code of 2026-09-17:  $E(10\text{ ms})/E(0)$ , or the time the run ended by element inversion. “Bounded, noisy” is the explicit state of Sec. 1.4.

| coupling                       | 1:0.9 II     | 1:0.9 EE       | 1:0.75 II    | 1:0.75 EE    | 1:0.5 II     | 1:0.5 EE       |
|--------------------------------|--------------|----------------|--------------|--------------|--------------|----------------|
| monolithic                     | 1.000        | 1.000          | 1.000        | 1.000        | 1.000        | 1.000          |
| overlap, Dirichlet             | 3.16         | fails 4.9 ms   | fails 1.5 ms | fails 2.1 ms | fails 0.9 ms | fails 1.3 ms   |
| nonoverlap, Dirichlet–Neumann  | fails 2.9 ms | 1.006          | fails 0.4 ms | 1.002        | fails 0.7 ms | 1.008          |
| overlap, impedance             | fails 7.2 ms | fails 8.4 ms   | fails 6.0 ms | 12.2         | fails 7.3 ms | 5.9            |
| nonoverlap, impedance (paired) | 1.000        | bounded, noisy | fails 5.0 ms | fails 2.6 ms | 0.43         | bounded, noisy |

## 6.7 The classical Robin–Robin condition

The Robin–Robin condition was run on both decompositions on 2026-09-18: **Schwarz RR nonoverlap** with the classical per-side transfer, and **Schwarz impedance overlap** with **impedance scale: 0**. Since the converged solution and the iteration both depend on  $\alpha$  on nonconforming meshes, two values, one near the Dirichlet limit and one near the Neumann limit, were used:  $\alpha = E/h = 10^{12} \text{ N/m}^3$  with  $h = 6.35 \text{ mm}$ , and  $\alpha = E/L = 2.7 \times 10^{10} \text{ N/m}^3$  with  $L = 254 \text{ mm}$ . Integrators, ratios, relaxation, and tolerances are those of Table 6. Every run converged in 2 to 4 Schwarz iterations per stop, so Table 7 shows the converged coupling.

Twenty-three of the 24 runs end by element inversion after exponential growth, between 0.2 and 7.4 ms; the one survivor (1:0.9, EE,  $\alpha = E/L$ ) stays within 5% to 10 ms. The growth is faster than with overlap Dirichlet transmission in 18 of the 24 cases, and the stiffer spring fails sooner in

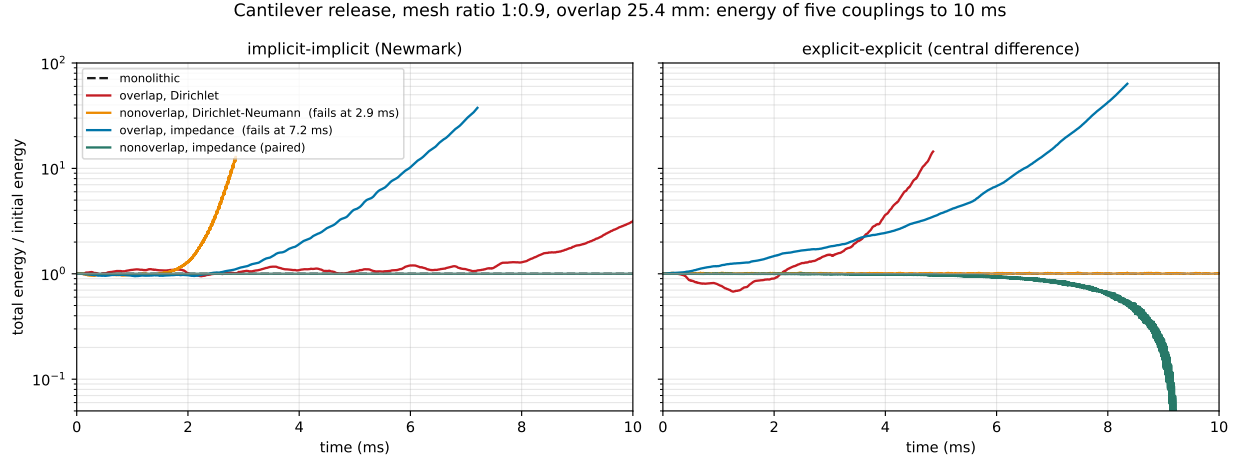


Figure 4: Cantilever at mesh ratio 1:0.9, overlap 25.4 mm, five couplings to 10 ms, code of 2026-09-17. Left: implicit–implicit. Right: explicit–explicit; the paired impedance curve ends in the bounded, noisy state.

Table 7: Cantilever, classical Robin–Robin coupling, overlap 25.4 mm:  $E(10 \text{ ms})/E(0)$ , or the time the run ended by element inversion.

| coupling                   | 1:0.9 II     | 1:0.9 EE         | 1:0.75 II    | 1:0.75 EE    | 1:0.5 II     | 1:0.5 EE     |
|----------------------------|--------------|------------------|--------------|--------------|--------------|--------------|
| overlap, $\alpha = E/h$    | fails 0.7 ms | fails 1.4 ms     | fails 0.8 ms | fails 1.9 ms | fails 0.4 ms | fails 0.8 ms |
| overlap, $\alpha = E/L$    | fails 0.9 ms | fails 1.1 ms     | fails 0.9 ms | fails 1.5 ms | fails 1.6 ms | fails 5.4 ms |
| nonoverlap, $\alpha = E/h$ | fails 1.2 ms | fails 2.4 ms     | fails 1.1 ms | fails 0.2 ms | fails 0.3 ms | fails 0.3 ms |
| nonoverlap, $\alpha = E/L$ | fails 7.1 ms | 0.96 (peak 1.05) | fails 7.4 ms | fails 4.5 ms | fails 3.4 ms | fails 0.9 ms |

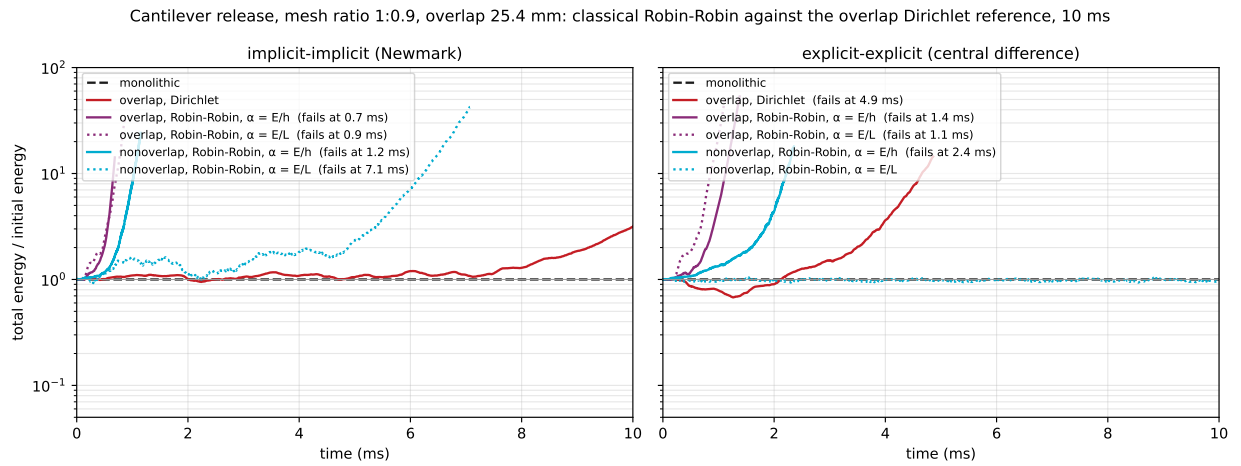


Figure 5: Cantilever at mesh ratio 1:0.9: classical Robin–Robin coupling on both decompositions at two Robin parameters, against the overlap Dirichlet reference, 10 ms.

10 of the 12 pairs. The converged coupled solution depends on  $\alpha$  (the two nonoverlap rows differ although both are converged to  $10^{-8}$ ): with per-side transfer on nonconforming meshes the discrete jump does not close, and the spring force on the residual jump scales with  $\alpha$ . The result is the one (7) predicts: a condition that reflects every wave has no term that damps the interface mode fed by the discrete mismatch, and the spring stores that mismatch. The dashpot of the impedance condition is the term that removes it.

## 7 Measurements on the nested cylinders

The cylinders (Sec. 1.3) were built as a second case that excites the instability independently of the beam: three materials with a fiftyfold impedance contrast, a compressive front instead of flexure, and a closed curved interface with caps, at a mild mesh ratio (1:0.8). Every subdomain is explicit. The Schwarz iteration uses Aitken relaxation for the Dirichlet–Neumann exchange, a fixed factor  $\theta = 0.5$  for the paired impedance and Robin–Robin couplings (Sec. 5.4), and no relaxation for the overlap couplings, with a relative tolerance of  $10^{-6}$  and an absolute tolerance of  $10^{-4}$ . The monolithic reference conserves the blended staggered energy to 0.01% over the  $10^4$  steps. Table 8 and Figs. 6–7 give the results.

Table 8: Nested cylinders, explicit on both sides, 5 ms:  $E(t)/E(0)$  at the times shown, or the time the run ended by element inversion. Robin–Robin at  $\alpha = E/h = 3.4 \times 10^{10}$  and  $\alpha = E/L = 2.8 \times 10^9 \text{ N/m}^3$ , with  $E$  the Young’s modulus of the filler,  $h = 6.35 \text{ mm}$ , and  $L = 76 \text{ mm}$ .

| coupling                                              | 0.5 ms                                       | 1 ms                        | 2 ms                        | 5 ms                       |
|-------------------------------------------------------|----------------------------------------------|-----------------------------|-----------------------------|----------------------------|
| overlap, Dirichlet                                    | 1.02                                         | 1.29                        | fails 1.54 ms at $7.4 E(0)$ |                            |
| overlap, impedance                                    | 1.02                                         | 1.76                        | fails 1.26 ms at $13 E(0)$  |                            |
| overlap, Robin–Robin, $\alpha = E/h$                  | 1.63                                         | fails 0.66 ms at $5.5 E(0)$ |                             |                            |
| overlap, Robin–Robin, $\alpha = E/L$                  | 1.53                                         | fails 0.75 ms at $11 E(0)$  |                             |                            |
| nonoverlap, Dirichlet–Neumann                         | 1.00                                         | 1.00                        | 1.01                        | 1.00                       |
| nonoverlap, Robin–Robin, $\alpha = E/h$               | 1.27                                         | fails 0.81 ms at $3.1 E(0)$ |                             |                            |
| nonoverlap, Robin–Robin, $\alpha = E/L$               | 1.08                                         | 1.26                        | 2.45                        | fails 3.59 ms at $11 E(0)$ |
| nonoverlap, impedance (paired), 2 iterations per stop | 0.79                                         | 0.54                        | 0.24                        | 0.065                      |
| same, 6 iterations per stop (to 1 ms)                 | 0.89                                         | 0.77                        | —                           | —                          |
| same, $\theta = 1$                                    | 0.997 to 0.3 ms, then the iteration diverges |                             |                             |                            |

**Dirichlet couplings.** Overlap Dirichlet coupling leaves the reference after 0.4 ms, grows exponentially, and ends by element inversion at 1.54 ms at  $7.4 E(0)$ : the same mechanism and a comparable rate as on the beam, although the interface here is closed and the meshes only mildly nonconforming. The nonoverlap Dirichlet–Neumann exchange (Aitken relaxation, 2 iterations per stop) stays within +3 and −1.3% for the whole 5 ms, as on the beam with explicit subdomains.

**Impedance couplings.** Overlap impedance coupling is unstable and faster than Dirichlet: it reaches  $13 E(0)$  at 1.26 ms. On the beam the same condition survived 6 to 8 ms; here the dashpot faces the inter-grid discrepancy of a compressive front resolved with 5 and 6 mm elements. The paired nonoverlap coupling never grows but dissipates strongly:  $0.54 E(0)$  at 1 ms and 0.065 at 5 ms. The interface velocity jump stalls at 0.1 to 0.14 of its initial size at every stop and is accepted by the displacement criterion, so the dashpot keeps absorbing it. Six iterations per stop keep  $0.77 E(0)$  at 1 ms instead of 0.54, so about half of the loss is iteration lag on a jump the criterion does not

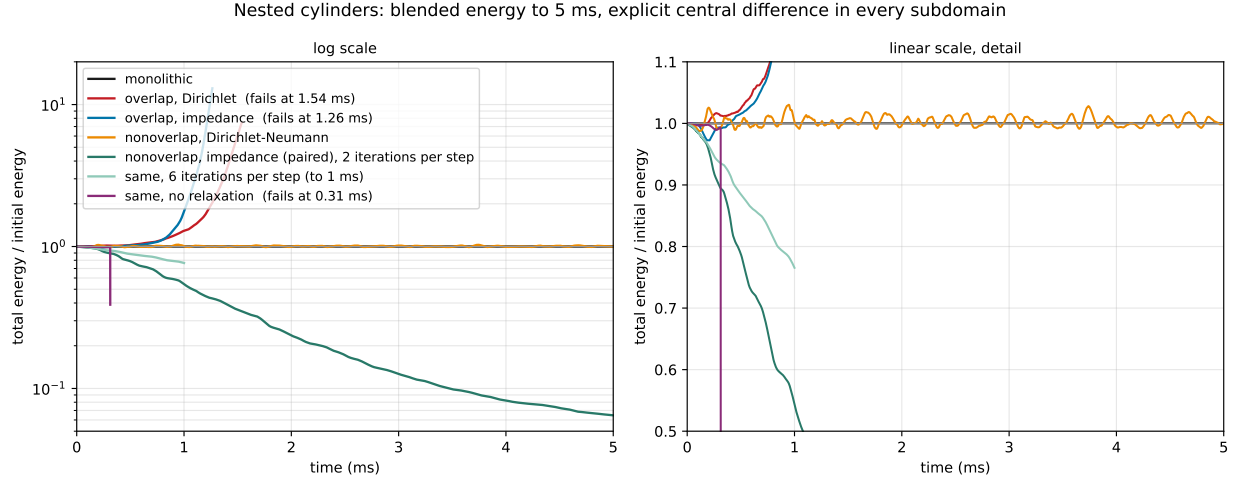


Figure 6: Nested cylinders, blended staggered energy to 5 ms: the Dirichlet and impedance couplings on both decompositions, with the two diagnostic variants of the paired coupling.

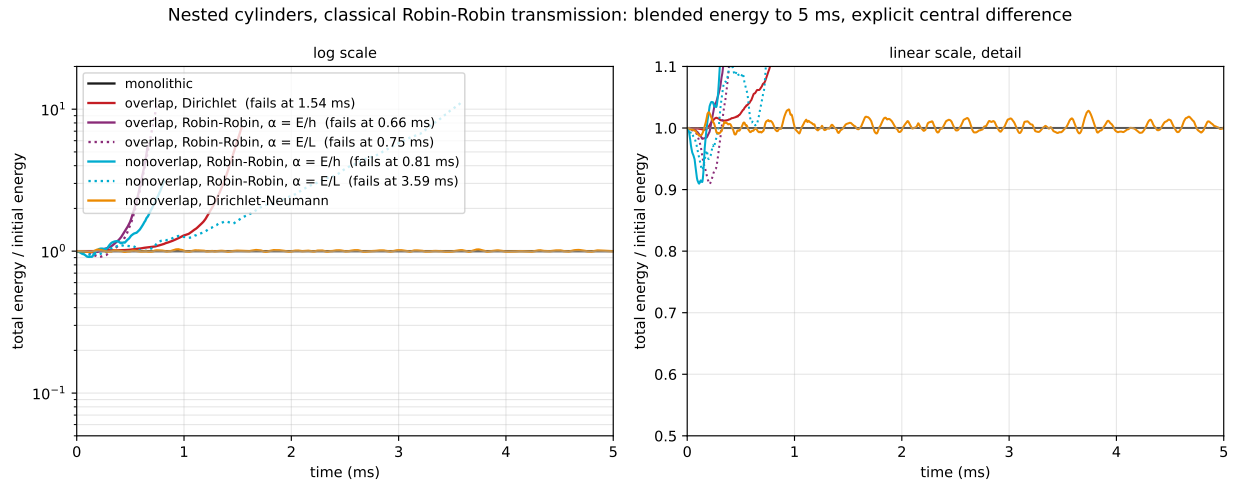


Figure 7: Nested cylinders, classical Robin–Robin coupling on both decompositions at two Robin parameters, against the overlap Dirichlet and nonoverlap Dirichlet–Neumann references.

measure; no relaxation keeps  $0.997 E(0)$  to 0.3 ms and then fails to converge within 256 iterations. Every Aitken form diverges on this pair (Sec. 5.4).

**Robin–Robin.** All four runs end by element inversion after exponential growth. The overlap runs fail sooner than overlap Dirichlet (1.54 ms) and overlap impedance (1.26 ms); the nonoverlap runs fail where the paired impedance coupling never grows and the Dirichlet–Neumann exchange holds. The Schwarz iteration converged in 1 to 2 iterations per stop. As on the beam, the stiffer spring fails sooner, and the softer nonoverlap spring lasts longest, 3.6 ms, still growing to  $11 E(0)$ .

## 8 The constrained formulation of the coupled problem

The measurements of Secs. 6–7 were made with transmission conditions chosen for the Schwarz iteration. This section states, from the literature, the coupled discrete problem that conserves energy by construction, locates where the nonoverlap Dirichlet–Neumann path of Norma departs from it, and lists the changes and the experiments that follow. It was written on September 30, 2026, after reading the four papers it rests on.

### 8.1 What the multiplier formulations prove

**Gravouil and Combescure** [4]. The subdomains are coupled in a dual Schur formulation: the equation of motion of subdomain  $k$  carries the interface force  $\mathbf{C}_k^T \Lambda$ , with  $\mathbf{C}_k$  the Boolean restriction to the interface (matching meshes) and  $\Lambda$  one vector of Lagrange multipliers, and the constraint is the continuity of the *velocity*,  $\sum_k \mathbf{C}_k \dot{\mathbf{u}}_k = 0$  (Eq. (10) there). With a coarse step  $\Delta T = m \Delta t$  on one side, the coarse velocity and the multiplier are interpolated linearly to the fine steps (Eqs. (12) and (13)). Theorem 1: if  $\gamma_k \geq 1/2$  in every subdomain, the coupling does not change the stability limit of any subdomain, for any pair of Newmark parameters, central difference with a lumped mass included. The interface work of Eq. (13) is a sum of  $m(m-1)/2$  negative squares (Eq. (49)): it is exactly zero at equal steps ( $m=1$ ), for every combination of Newmark schemes, and nonpositive otherwise, zero only when the coarse velocity is linear over the coarse step. The energy that velocity continuity conserves is the Newmark discrete energy of the system differentiated in time,  $\mathbf{M} \dot{\mathbf{a}} + \mathbf{K} \mathbf{v} = \dot{\mathbf{f}}$ , summed over the subdomains,

$$E_2 = \sum_k \left( \frac{1}{2} \mathbf{a}_k^T \mathbf{A}_k \mathbf{a}_k + \frac{1}{2} \mathbf{v}_k^T \mathbf{K}_k \mathbf{v}_k \right), \quad \mathbf{A}_k = \mathbf{M}_k + \frac{1}{2} \Delta t_k^2 (2\beta_k - \gamma_k) \mathbf{K}_k \quad (19)$$

(Prakash and Hjelmstad [6], Eqs. (56)–(58) and (71)), with  $\mathbf{a}_k$  and  $\mathbf{v}_k$  the acceleration and the velocity of subdomain  $k$ ;  $\mathbf{A}_k$  is  $\mathbf{M}_k$  for average acceleration and  $\mathbf{M}_k - \frac{1}{4} \Delta t_k^2 \mathbf{K}_k$  for central difference.  $E_2$  is not the physical energy  $E_1 = \sum_k (\frac{1}{2} \mathbf{v}_k^T \mathbf{M}_k \mathbf{v}_k + \text{SE}(\mathbf{u}_k))$  that the blended output computes. Continuity of the displacement with average acceleration conserves  $E_1$  exactly. For identical integrators at equal steps both are conserved. The interface operator (Eqs. (19), (22), and (25) there) is

$$\mathbf{H} = \sum_k \gamma_k \Delta t_k \mathbf{C}_k \tilde{\mathbf{M}}_k^{-1} \mathbf{C}_k^T, \quad \tilde{\mathbf{M}}_k = \mathbf{M}_k + \beta_k \Delta t_k^2 \mathbf{K}_k, \quad (20)$$

diagonal for an explicit subdomain with a lumped mass; the multiplier at each fine step solves  $\mathbf{H} \Lambda_j = -\sum_k \mathbf{C}_k \dot{\mathbf{u}}_{k,j}^{\text{free}}$  from the velocities of the subdomains advanced without interface forces (Eq. (24)). The nonlinear extension (Sec. 3 there) iterates the implicit subdomain with the interface, the multiplier increment following the free implicit velocity increment (Eq. (78)). Measured dissipation with subcycling: 1.3% at a ratio of 10 000 on a linear beam, 0.4% at 1 000 with plasticity, 1.9% at 100 on a stiffened pipe.

**Prakash and Hjelmstad [6].** The velocity constraint is imposed only at the end of the coarse step (Eq. (25) there); the fine side receives the multiplier interpolated linearly plus the unbalanced reaction  $\mathbf{S}_j$  of the coarse subdomain's free solution at the intermediate steps (Eq. (40)). The interface problem is solved once per coarse step,  $\mathbf{H}\Lambda_m = \mathbf{C}_A\dot{\mathbf{Y}}_m^A + \mathbf{C}_B\dot{\mathbf{Y}}_m^B$  with  $\mathbf{H} = \mathbf{C}_A\dot{\mathbf{Y}}_m^A + \mathbf{C}_B\dot{\mathbf{Y}}_m^B$  (Eq. (52)), where  $\mathbf{Y}$  is the response of each subdomain to a unit load at one interface degree of freedom, one coarse step from rest for the coarse side and a load ramped over the  $m$  fine steps for the fine side (their Sec. 4.1.2), computed once when the problem is linear and the steps constant. The energy change is exactly zero for every integer ratio and every pair of Newmark schemes (Eq. (70)); the proof uses linear elasticity. The initial acceleration must be computed on the undecomposed system (Eq. (54)).

**Mahjoubi, Gravouil, and Combescure [7].** The velocity constraint and the free/link decomposition extended to the Simo and Krenk schemes, with  $\mathbf{H}$  factorized once. Remark 2 there: only the velocity is continuous; the displacement and the acceleration are not imposed, and the displacement gap grows in time (drift). Sec. 3.5.2: with different time scales the interpolated coupling is first-order accurate in time, the Prakash–Hjelmstad coupling second-order. Sec. 4.3 is an explicit–explicit case with incompatible meshes and a time step ratio of 10: the fine interface nodes are constrained to follow the coarse interpolation, the velocity jump is  $10^{-14}$ , the displacement drift below 1%, and the dissipation below 1% of the external work over 10 000 steps. The nonconforming treatment of this line of work is Herry, Di Valentin, and Combescure [9]; the nonconforming asynchronous conserving method is Subber and Matouš [10]. The review of the family is Gravouil, Combescure, and Brun [8].

**Dvořák et al. [20].** Localized Lagrange multipliers [19] with a frame between the subdomains, continuity of the *acceleration* (Eq. (19) there), and an a posteriori mapping of the frame displacement and velocity onto the interface nodes (Eqs. (36) and (37)), so that all three fields are continuous and the interface work is exactly zero. Arbitrary, non-integer, time-varying steps; explicit in structure; any number of subdomains. The total energy is not exactly conserved: the error appears in the subdomain energies when a discontinuity crosses the interface and vanishes as the steps decrease (their Sec. 7.1). Each subdomain must expose an interface region, a layer around the interface integrated at the partner's time levels (their Sec. 5), and the formulation is given for linear elasticity.

## 8.2 The pseudo-energy identity of the subcycled exchange

Connors, Owen, Kuberry, and Bochev [5] state the energy balance per side. For side  $k$  advanced by Newmark steps  $\Delta t_k$  with a linear stiffness  $\mathbf{K}_k$ , an interface force  $\mathbf{f}_k$  on its interface rows, and no other loads, the pseudo-energy

$$E_{2,k}^n = \frac{1}{2} (\mathbf{a}_k^n)^T \mathbf{S}_k \mathbf{a}_k^n + \frac{1}{2} (\mathbf{v}_k^n)^T \mathbf{K}_k \mathbf{v}_k^n, \quad \mathbf{S}_k = \mathbf{M}_k + \left(\beta_k - \frac{\gamma_k}{2}\right) \Delta t_k^2 \mathbf{K}_k, \quad (21)$$

which is  $E_2$  of (19), changes over the  $m_k$  steps of a window by (their Eq. (102))

$$E_{2,k}^{m_k} - E_{2,k}^0 + \left(\gamma_k - \frac{1}{2}\right) \sum_{n=1}^{m_k} (\Delta \mathbf{a}_k^n)^T \mathbf{S}_k \Delta \mathbf{a}_k^n = \frac{1}{\Delta t_k} \sum_{n=1}^{m_k} (\Delta \mathbf{v}_k^n)^T \mathbf{C}_k^T \Delta \mathbf{f}_k^n, \quad (22)$$

with  $\Delta(\cdot)^n = (\cdot)^n - (\cdot)^{n-1}$ . With  $\gamma_k = 1/2$  the second term vanishes, and  $E_{2,k}$  is a norm when  $\mathbf{S}_k$  is positive definite, which for central difference ( $\beta = 0$ ) is its stability limit.

Let the Dirichlet side  $D$  take one step  $\Delta T$  and the Neumann side  $N$  take  $m$  steps  $\Delta T/m$ , the interface force on  $D$  be  $\lambda$  (the negative of its d'Alembert reaction), and the force on  $N$  at its substep  $j$  be  $-\Pi_D^T[(1 - j/m)\lambda^0 + (j/m)\lambda^1]$ , the linear interpolation that the exchange produces (Connors et al. Eqs. (89) and (90)). Then  $\Delta \mathbf{f}_N^j = -\Pi_D^T \Delta \lambda / m$  for every substep, and the right side of (22) for  $N$  is

$$\frac{m}{\Delta T} \sum_{j=1}^m (\Delta \mathbf{v}_N^j)^T \left( -\Pi_D^T \frac{\Delta \lambda}{m} \right) = -\frac{1}{\Delta T} (\Pi_D (\mathbf{v}_N^m - \mathbf{v}_N^0))^T \Delta \lambda, \quad (23)$$

while that of  $D$  is  $\Delta T^{-1}(\mathbf{v}_D^1 - \mathbf{v}_D^0)^T \Delta \lambda$ . If  $\mathbf{v}_D = \Pi_D \mathbf{v}_N$  at both ends of the window, the two cancel and  $E_{2,D} + E_{2,N}$  is constant; this is Connors et al. Eq. (103) with  $\Pi_D$  in place of their Boolean interface map, their condition (108) at  $r = 1$ , and Prakash and Hjelmstad's Eq. (70). The conditions are: a linear stiffness on both sides;  $\gamma = 1/2$  on both sides (otherwise each side dissipates by the second term of (22));  $\mathbf{S}_k$  positive definite, for  $E_2$  to bound the solution; the force on  $N$  at the window start equal to the converged reaction of the previous window, which holds in Norma because `restore_stop_state` runs before `subcycle` pushes the anchor snapshot; and loads on the interface rows of  $D$  constant within the window, because the reaction is formed from their current value. An explicit member keeps its own stability limit.

The same sum cancels when the fine side is the Dirichlet side and meets the linearly interpolated coarse velocity at every fine step, the arrangement of Gravouil and Combescure (their Eqs. (21)–(24) solve the multiplier at every fine step):  $\Delta \mathbf{v}_D^j = \Pi_D \Delta \mathbf{v}_N / m$  at each substep, so the Dirichlet sum is  $\Delta T^{-1}(\Pi_D \Delta \mathbf{v}_N)^T (\lambda^m - \lambda^0)$ , which cancels the coarse side's term. What that arrangement does not have is the flux projection property of Connors et al. (Eq. (80)): the impulse that the fine side receives over the window, the trapezoid sum of its reactions at every substep, is not the impulse that the coarse side receives, the trapezoid of the two end values. Sec. 8.5 measures that impulse residual.

### 8.3 Where the Dirichlet–Neumann path departs from it

From `src/schwarz.jl` and `src/simulation.jl` at commit 3e4f3b7c:

1. The force transmitted to the Neumann side is the static reaction  $-\mathbf{f}_{\text{int}}$  of the partner (`get_dst_force`); the d'Alembert reaction  $\mathbf{M}\mathbf{a} + \mathbf{f}_{\text{int}} - \mathbf{f}_{\text{body}}$  is added only for the impedance condition with adjoint pairing. In the multiplier formulations  $\Lambda$  is the full dynamic reaction of either side. This is the omission that Sec. 6.4 measured on the impedance path as a drain of 9.6% per millisecond on conforming meshes, and it is present on every Dirichlet–Neumann run, including the explicit ones that conserve within 1 to 3%.
2. The two transfer operators are built separately, in `compute_dirichlet_projector` and `compute_neumann_projector`:  $\Pi_D = \mathbf{W}_{\text{dst}}^{-1} \mathbf{L}$  on the Dirichlet side and  $\Pi_N = \mathbf{L} \mathbf{H}_{\text{src}}^{-1}$  on the Neumann side, each with its own choice of the side on which  $\mathbf{L}$  is integrated. They satisfy (15) only when the same cross matrix is chosen on both sides; the impedance path builds both from one matrix (the function `compute_paired_impedance_schwarz_projectors!`) and the Dirichlet–Neumann path does not.
3. The Dirichlet side imposes the displacement, the velocity, and the acceleration together (`coupling_weak dbc`). At equal steps and at convergence this contains the velocity constraint; under subcycling the three interpolated fields are not consistent with the Newmark relations of the receiving side, and the velocity alone is what the theorem constrains.
4. The convergence criterion (`update_schwarz_convergence_criterion`) is the change of  $\mathbf{u} + \Delta t \dot{\mathbf{u}}$  over all degrees of freedom of all subdomains. It measures neither the velocity jump on

the interface nor the balance of the interface forces, which is why a converged iteration leaves a jump of 0.10 to 0.14 on the cylinders (Sec. 7) and why the converged solution depends on the relaxation factor (Sec. 9.3, item (ii)).

5. The initial acceleration is computed subdomain by subdomain with the Schwarz degrees of freedom excluded (`initialize` in `src/time_integrator.jl`), against Eq. (54) of Prakash and Hjelmstad.

The Dirichlet–Neumann fixed point is therefore not the constrained problem, and the theorems of Sec. 8.1 do not apply to it; its conservation with explicit subdomains and its failure with implicit ones (Sec. 9, finding 8) are properties of a different scheme.

## 8.4 Changes and reference experiments

The framework to adopt is the constrained problem of Gravouil and Combescure, with the Schwarz iteration as one of its solvers. In order:

1. On the nonoverlap path, transmit the d’Alembert reaction, build both transfer operators from one cross matrix so that  $\Pi_N = \Pi_D^T$ , and impose the velocity constraint; the displacement and the acceleration then follow from the receiving side’s Newmark relations, or are imposed only at equal steps.
2. Stop the iteration on the projected velocity jump and on the interface force residual.
3. Compute the initial acceleration on the coupled system, by a Schwarz iteration on the acceleration at  $t = 0$  or by a monolithic solve.
4. For explicit subdomains, replace the iteration by the direct interface solve:  $\mathbf{H}$  of (20) is  $\gamma\Delta t$  times the inverse of the lumped mass on the interface, one division per interface degree of freedom per step.
5. For implicit subdomains, keep the Dirichlet–Neumann iteration with Aitken relaxation, converged on the constraint, or solve the interface problem by conjugate gradients with one solve of  $\bar{\mathbf{M}}$  per subdomain per iteration, the operation of the transient FETI method [11, 12].
6. For different time steps, make the coarse-step side the Dirichlet side: it imposes the fine side’s velocity at the end of the window, and the fine side receives the coarse side’s d’Alembert reaction interpolated linearly between the window’s end values. This is the  $r = 1$  scheme of Connors et al. [5] (their Eqs. (53)–(55), (89)–(90), and (104)–(105) with  $k = 1$ ), whose reaction is their flux  $\Sigma^N$ , and the scheme of Prakash and Hjelmstad (their Eqs. (25), (40), and (70)); its interface terms cancel at the fixed point of the iteration (Sec. 8.2), so it conserves  $E_2$  without a direct interface solve.

The Dirichlet side of each pair is the implicit member, and for two members with the same integrator on a nonconforming interface the finer side. The first runs of the reference experiments assigned it to the subdomain with the smaller interface tributary mass, the explicit member of a mixed pair, which the gain analysis of Sec. 8.5 shows to be divergent. Changes 1 to 3 are implemented and measured there.

The reference experiments, with the outcome the theorems require:

- Beam, matching meshes, equal steps, each of II, IE, EI, and EE, after changes 1 to 3, converged to  $10^{-12}$  on the constraint:  $E_2$  of (19) under the velocity constraint, and  $E_1$  under the displacement constraint with average acceleration on both sides, is constant to roundoff over 10 ms. Any residual measures what remains of the departure.

- The same at ratio 1:0.75: the identity holds with the projected velocities, so the residual is the quadrature error of the cross matrix, zero for nested facets and small otherwise; it must not grow in time.
- The same with a 4:1 step ratio: dissipation, nonpositive, of the order of one percent, and zero for explicit–explicit subdomains when the coarse step is the critical step (Sec. 4.4 of Gravouil and Combescure).

**Against the requirements of a coupling framework.** The constrained problem gives the guarantees, the nonconforming meshes through the projected constraint, the heterogeneous integrators, and the different steps. Integration into an existing code needs, beyond the boundary conditions the Schwarz iteration already uses, one operation per subdomain: a solve with the effective dynamic matrix  $\tilde{\mathbf{M}}$  for a given interface load, which an implicit code has as a Newton iteration at fixed tangent and an explicit code has as a division by the lumped mass. The overlap decomposition has no place in this framework (Sec. 4), and every measured overlapping case failed; it should be retired for dynamics.

## 8.5 Measurements of the constrained exchange

Changes 1 to 3 are implemented for equal time steps (**constrained: true** on both sides of a **Schwarz DN nonoverlap** pair, with **constraint: velocity** or **displacement**). The cantilever runs below use the time step  $10^{-6}$  s, 10 ms, a relative tolerance of  $10^{-12}$  on the jump of the constrained quantity and on the interface force residual, at most 256 Schwarz iterations per stop, and the direct linear solver. Energies are  $E_1$ , the blended physical energy of Sec. 1.4, and  $E_2$  of (19). The jump is the one-sided relative jump  $\|\mathbf{u}_D - \Pi_D \mathbf{u}_N\|_{\mathbf{W}_D} / \|\mathbf{u}_D\|_{\mathbf{W}_D}$  on the Dirichlet side  $D$ , with  $N$  the Neumann side and  $\mathbf{W}_D$  the boundary mass matrix of  $D$ .

**Ground truth.** An undecomposed run of the same beam at  $\Delta t = 5 \times 10^{-7}$  s keeps  $E_1$  and  $E_2$  within  $1.7 \times 10^{-13}$  of their initial values over 10 ms with average acceleration and within  $2.7 \times 10^{-14}$  with central difference. The present Dirichlet–Neumann exchange at the same step, tolerance  $10^{-8}$ , and  $\theta = 0.5$  inverts an element at 0.46 ms with implicit subdomains ( $E_1/E_1(0) = 1.91$  and  $E_2/E_2(0) = 34$  at 0.4 ms) and ends at  $E_1/E_1(0) = 1.013$  and  $E_2/E_2(0) = 1.006$  at 10 ms with explicit ones.

**The force residual of the present exchange.** The relative difference between the force applied on the Neumann side and the transposed projection of the d’Alembert reaction of the Dirichlet side, measured at the end of every stop in the inverse norm of the Neumann side’s boundary mass matrix, has the median 0.83 (maximum 4.2) with explicit subdomains and 1.47 (maximum 4.6) with implicit ones: the static reaction  $-\mathbf{f}^{\text{int}}$  that the present exchange transmits misses an interface inertia of the size of the reaction itself. This is departure 1 of Sec. 8.3, measured directly. Under the constrained exchange the same residual is below  $4 \times 10^{-16}$ .

**The Dirichlet role.** The gain of one Schwarz iteration of the constrained exchange, linearized at a stop, is

$$\mathbf{G} = -\Pi_D \mathbf{H}_N \Pi_D^T \mathbf{S}_D, \quad (24)$$

with  $\mathbf{H}_N = \mathbf{C}_N \tilde{\mathbf{M}}_N^{-1} \mathbf{C}_N^T$  the interface flexibility of the Neumann side ( $\tilde{\mathbf{M}} = \mathbf{M} + \beta \Delta t^2 \mathbf{K}$ , the lumped mass for central difference) and  $\mathbf{S}_D$  the interface dynamic stiffness of the Dirichlet side ( $\mathbf{H}_D^{-1}$  for Newmark, the lumped interface mass for central difference). Fixed relaxation  $\theta$  converges

if and only if the spectral radius of  $(1-\theta)\mathbf{I}+\theta\mathbf{G}$  is below one. Table 9 lists the computed eigenvalues. On conforming meshes the two sides have the same lumped tributary masses, so these do not select the Dirichlet side; the consistent-mass flexibility of an implicit side is up to 13 times the inverse lumped mass of an explicit side, and with the explicit member as the Dirichlet side every  $\theta \geq 0.5$  diverges. All four runs of that assignment (IE and EI,  $\theta \in \{0.5, 0.75, 1\}$ ) inverted an element at the first stop; with recursive Aitken they stopped on a stalled residual of  $2 \times 10^{-9}$  and  $2 \times 10^{-10}$ . The rule is: *the implicit member is the Dirichlet side*; for two members with the same integrator on a nonconforming interface, the finer side.

Table 9: Eigenvalues of the gain (24) at  $\Delta t = 10^{-6}$  s and the spectral radius  $\rho$  of the relaxed iteration. Conforming: the meshes of the cantilever example; 1:0.5 and 1:0.75: free part 6.35 mm, clamped part 12.7 and 8.47 mm. D is the Dirichlet side.

| meshes     | pair, D              | eigenvalues       | $\rho(0.5)$ | $\rho(0.75)$ | $\rho(1)$ |
|------------|----------------------|-------------------|-------------|--------------|-----------|
| conforming | II, either           | $[-1.09, -0.92]$  | 0.05        | 0.57         | 1.09      |
| conforming | EE, either           | $-1$              | 0           | 0.50         | 1.00      |
| conforming | IE or EI, D explicit | $[-13.3, -1.63]$  | 6.1         | 9.7          | 13.3      |
| conforming | IE or EI, D implicit | $[-0.61, -0.075]$ | 0.46        | 0.21         | 0.61      |
| 1:0.5      | II, D fine           | $[-0.55, 0]$      | 0.50        | 0.25         | 0.55      |
| 1:0.5      | II, D coarse         | $[-1.99, -1.83]$  | 0.50        | 1.24         | 1.99      |
| 1:0.5      | EE, D fine           | $[-0.50, 0]$      | 0.50        | 0.25         | 0.50      |
| 1:0.5      | EE, D coarse         | $[-8.0, -2.0]$    | 3.5         | 5.8          | 8.0       |
| 1:0.75     | II, D fine           | $[-0.82, 0]$      | 0.50        | 0.37         | 0.82      |
| 1:0.75     | EE, D fine           | $[-0.75, 0]$      | 0.50        | 0.31         | 0.75      |

**Relaxation.** For two members with the same integrator on conforming meshes the eigenvalues cluster at  $-1$ , so  $\theta = 0.5$  is optimal (7 iterations per stop for II, 1 for EE),  $\theta = 0.75$  contracts by 0.5 per iteration (39 and 35 iterations), and  $\theta = 1$  does not converge (element inversion, or 256 iterations at the first stop for EE, where the iterate alternates between two states). With the fine side as the Dirichlet side the eigenvalues fill  $[g, 0]$ , and the zero eigenvalues belong to the components of the fine trace that the coarse trace cannot represent; they contract by  $1 - \theta$ . The recursive Aitken factor does not settle where the eigenvalues cluster at  $-1$ : on the implicit pair it alternates between 0.5 and 0.998, the iteration stalls for one iteration at a relative velocity jump of  $5.9 \times 10^{-10}$ , and the stall rule (a decrease below 5% in one iteration once the displacement update has converged, with **stalled interface jump action: abort**) ended the run at the first stop. It also ended the IE run with the implicit Dirichlet side at 7.7 ms ( $2.7 \times 10^{-11}$ ). The Aitken residual is formed on the whole velocity field of the Neumann side, not on its interface trace.

**The first step and the coupled initial acceleration.** With the initial acceleration computed per subdomain and the interface held fixed (Sec. 8.3, departure 5), the first step changed  $E_2$  by 0.18 to 0.9% and  $E_1$  by  $5 \times 10^{-5}$  (II) and  $5 \times 10^{-4}$  (EE), and conservation held only from the first step on. In the explicit pair the difference of the two interface accelerations then alternates in sign at every step: velocity continuity fixes the sum of two consecutive accelerations on each side, so an initial difference is never removed, the displacement jump alternates between  $2 \times 10^{-3}$  and  $10^{-14}$ , and  $E_1$  alternates by  $10^{-3}$ . The remedy is change 3: a Schwarz iteration at  $t = 0$  in which the Dirichlet side imposes the projected partner acceleration, the Neumann side receives the d’Alembert reaction, and each side recomputes its acceleration with the interface rows included. It converges in 2 iterations on conforming meshes with the same integrator, 30 for the mixed pairs,

and 16 to 24 on the nonconforming meshes, and inverts no element; the inversion recorded for the trust-Schwarz re-solve of the present exchange does not occur.

Table 10: Constrained exchange on the conforming cantilever, completed runs.  $E(10)$  is the ratio to the value at  $t = 0$  at 10 ms;  $\delta E = \max_t |E(t)/E(0) - 1|$ ;  $\delta_1 E_2$  is the same from the first step on; jump is the largest displacement jump over the last 100 stops; it. is the mean and maximum number of Schwarz iterations per stop; wall time in seconds on four threads. “Dir. impl.” marks the implicit member as the Dirichlet side; the free side is the Dirichlet side otherwise.

| pair, constraint                          | $\theta$ | $E_1(10)$ | $E_2(10)$ | $\delta E_1$ | $\delta E_2$ | $\delta_1 E_2$ | jump    | it.      | wall |
|-------------------------------------------|----------|-----------|-----------|--------------|--------------|----------------|---------|----------|------|
| <i>initial acceleration per subdomain</i> |          |           |           |              |              |                |         |          |      |
| II, displacement                          | 0.5      | 1.000051  | 0.994821  | 5.1e−5       | 5.2e−3       | 2.0e−7         | 5.5e−13 | 7.1/8    | 172  |
| II, displacement                          | 0.75     | 1.000051  | 0.994823  | 5.1e−5       | 5.2e−3       | 2.1e−6         | 1.0e−12 | 38.9/43  | 784  |
| II, displacement                          | Aitken   | 1.000051  | 0.994820  | 5.1e−5       | 5.2e−3       | 9.2e−7         | 9.9e−13 | 10.4/155 | 269  |
| II, velocity                              | 0.5      | 1.000051  | 0.994821  | 5.1e−5       | 5.2e−3       | 6.8e−11        | 1.7e−12 | 7.0/8    | 206  |
| II, velocity                              | 0.75     | 1.000051  | 0.994821  | 5.1e−5       | 5.2e−3       | 9.9e−12        | 2.1e−11 | 38.7/41  | 714  |
| EE, velocity                              | 0.5      | 0.999189  | 1.001776  | 1.0e−3       | 1.8e−3       | 4.1e−14        | 2.0e−3  | 1.0/1    | 46   |
| EE, velocity                              | 0.75     | 0.999189  | 1.001776  | 1.0e−3       | 1.8e−3       | 1.3e−10        | 2.0e−3  | 34.6/36  | 190  |
| EE, velocity                              | Aitken   | 0.999189  | 1.001776  | 1.0e−3       | 1.8e−3       | 4.1e−14        | 2.0e−3  | 1.0/1    | 48   |
| IE, velocity, Dir. impl.                  | 0.5      | 0.999489  | 0.991040  | 1.4e−3       | 9.0e−3       | 2.7e−11        | 1.7e−3  | 28.9/32  | 305  |
| IE, velocity, Dir. impl.                  | 0.75     | 0.999489  | 0.991040  | 1.4e−3       | 9.0e−3       | 6.2e−11        | 1.7e−3  | 18.0/19  | 236  |
| IE, velocity, Dir. impl.                  | 1        | 0.999489  | 0.991040  | 1.4e−3       | 9.0e−3       | 4.3e−13        | 1.7e−3  | 53.1/56  | 574  |
| EI, velocity, Dir. impl.                  | 0.5      | 1.003183  | 0.995852  | 3.5e−3       | 4.1e−3       | 4.1e−10        | 1.8e−3  | 27.1/28  | 430  |
| EI, velocity, Dir. impl.                  | 0.75     | 1.003183  | 0.995852  | 3.5e−3       | 4.1e−3       | 5.0e−11        | 1.8e−3  | 14.2/15  | 225  |
| EI, velocity, Dir. impl.                  | 1        | 1.003183  | 0.995852  | 3.5e−3       | 4.1e−3       | 1.5e−10        | 1.8e−3  | 43.1/46  | 491  |
| EI, velocity, Dir. impl.                  | Aitken   | 1.003183  | 0.995852  | 3.5e−3       | 4.1e−3       | 4.2e−11        | 1.8e−3  | 12.7/15  | 167  |
| <i>coupled initial acceleration</i>       |          |           |           |              |              |                |         |          |      |
| II, displacement                          | 0.5      | 1.000000  | 1.000000  | 2.2e−11      | 5.3e−11      | 5.3e−11        | 5.4e−13 | 7.1/8    | 215  |
| II, velocity                              | 0.5      | 1.000000  | 1.000000  | 2.8e−11      | 7.0e−11      | 7.0e−11        | 1.7e−12 | 7.0/8    | 206  |
| EE, velocity                              | 0.5      | 1.000000  | 1.000000  | 1.2e−14      | 3.6e−14      | 3.8e−14        | 7.5e−15 | 1.0/1    | 46   |
| IE, velocity, Dir. impl.                  | 0.5      | 0.999638  | 1.000000  | 1.2e−3       | 6.3e−12      | 6.4e−12        | 9.6e−4  | 29.0/32  | 417  |
| EI, velocity, Dir. impl.                  | 0.5      | 1.003366  | 1.000000  | 3.7e−3       | 4.2e−10      | 4.2e−10        | 9.3e−4  | 27.1/28  | 419  |

**What Table 10 shows.** With the coupled initial acceleration, the energy that the constraint matches is constant over 10 ms to between  $4 \times 10^{-14}$  and  $4 \times 10^{-10}$ :  $E_2$  under the velocity constraint for every pair, and both  $E_1$  and  $E_2$  for the identical integrators. The residual is set by the tolerance  $10^{-12}$  accumulated over  $10^4$  stops, not by roundoff (undecomposed:  $10^{-13}$ ); the explicit pair at  $\theta = 0.5$  converges in one iteration exactly and reaches  $4 \times 10^{-14}$ . The final energies of runs that differ only in  $\theta$  agree to between  $10^{-12}$  and  $5 \times 10^{-10}$ , the same tolerance effect. Under the displacement constraint with the per-subdomain initial acceleration  $E_2$  drifted by  $2 \times 10^{-7}$  to  $2 \times 10^{-6}$ , because the acceleration  $(\mathbf{u} - \mathbf{u}^{\text{pre}})/(\beta \Delta t^2)$  multiplies the displacement iteration error by  $4/\Delta t^2$ ; with the coupled initial acceleration that drift is  $5 \times 10^{-11}$ . For the mixed pairs  $E_1$  is not conserved (0.4%), which is the expected behavior of the velocity constraint between different integrators. The displacement drift of Mahjoubi et al. appears in the mixed pairs only: its root mean square over the interface is between  $7 \times 10^{-6}$  and  $1.1 \times 10^{-5}$  m in every millisecond window of the 10 ms (0.1 to 0.2% of the initial interface displacement  $6.3 \times 10^{-3}$  m), and it grows by at most 10% from the first to the last window. In these runs and in II under the velocity constraint the root mean square velocity jump reaches the absolute floor  $10^{-15}$  m/ $10^{-6}$  s =  $10^{-9}$  m/s, at which the stopping rule also accepts the iterate.

**Nonconforming meshes.** The one-sided identity  $\mathbf{f}_N^T \dot{\mathbf{u}}_N = \mathbf{r}_D^T \Pi_D \dot{\mathbf{u}}_N = \mathbf{r}_D^T \dot{\mathbf{u}}_D$  holds for any cross matrix  $\mathbf{B}$ , so the matched energy should be conserved for any mesh ratio. Free part 6.35 mm, clamped part 12.7 mm (ratio 1:0.5, nested) and 8.47 mm (1:0.75, not nested), fine side as the Dirichlet side,  $\theta = 0.5$ , coupled initial acceleration:

| ratio  | pair, constraint | $E_1(10)$ | $E_2(10)$ | $\delta E_1$ | $\delta E_2$ | jump    | it.     | wall |
|--------|------------------|-----------|-----------|--------------|--------------|---------|---------|------|
| 1:0.5  | II, displacement | 1.000000  | 1.000000  | 1.2e-10      | 4.3e-11      | 9.8e-13 | 17.3/19 | 437  |
| 1:0.5  | II, velocity     | 1.000000  | 1.000000  | 1.2e-10      | 1.8e-11      | 5.6e-12 | 17.2/18 | 436  |
| 1:0.5  | EE, velocity     | 1.000000  | 1.000000  | 3.5e-11      | 6.9e-11      | 1.7e-11 | 23.7/26 | 169  |
| 1:0.75 | II, velocity     | 1.000000  | 1.000000  | 1.7e-11      | 5.2e-11      | 1.5e-12 | 13.5/14 | 350  |
| 1:0.75 | EE, velocity     | 1.000000  | 1.000000  | 2.2e-11      | 6.1e-11      | 6.0e-12 | 16.2/17 | 157  |

All five runs complete and conserve to the same level as the conforming ones. The certificates of  $\mathbf{B}$  are at roundoff at 1:0.5 (partition of unity of both projectors  $10^{-15}$ ,  $|\mathbf{B}\mathbf{1} - \mathbf{W}_D\mathbf{1}|$  and  $|\mathbf{B}^T\mathbf{1} - \mathbf{W}_N\mathbf{1}|$   $10^{-16}$  relative). At 1:0.75 the projector onto the coarse side reproduces constants only to  $1.2 \times 10^{-2}$  and  $\mathbf{B}^T\mathbf{1}$  differs from  $\mathbf{W}_N\mathbf{1}$  by  $3.1 \times 10^{-3}$  relative, the quadrature error of a coarse basis function that kinks inside a fine facet; the energy is still conserved to  $6 \times 10^{-11}$ , because with the fine side as the Dirichlet side only  $\Pi_D$ , whose certificate is  $7 \times 10^{-16}$ , and its transpose enter the exchange. At this ratio the paired impedance coupling inverted an element at 2.6 to 5 ms (Sec. 6.4); the constrained exchange completes 10 ms.

**Different time steps.** The velocity constraint admits different steps on the two sides: the side with the finer step receives the partner's velocity (Dirichlet side) or d'Alembert reaction (Neumann side) interpolated linearly in time from the partner's substep history, and the stopping rule takes the largest jump and force residual over the substeps of the stop. Reference experiment 3 is the beam with one side subcycled 4:1 (coarse step  $10^{-6}$  s, fine step  $2.5 \times 10^{-7}$  s),  $\theta = 0.5$ , relative tolerance  $10^{-12}$  with no absolute tolerance, **stalled interface jump action: warn**, 10 ms (Table 11). The energy output carries the interface impulse residual of each window: the trapezoid sum over the Neumann side's substeps of the force it received, minus  $\Pi_D^T$  times the trapezoid sum of the Dirichlet side's reaction over its own substeps, summed over the interface nodes per component (net, Ns) and in the  $\mathbf{W}_N^{-1}$  norm relative to the transferred impulse.

On the conforming beam, with the coarse-step (clamped) side as the Dirichlet side, the arrangement of Sec. 8.2,  $E_2$  stays within  $4.5 \times 10^{-13}$  (EE) and  $1.6 \times 10^{-12}$  (II),  $E_1$  within 0.11% and 0.38%, and the impulse residual is at roundoff (net at most  $2.6 \times 10^{-14}$  Ns, relative at most  $1.5 \times 10^{-12}$ ). With the fine-step (free) side as the Dirichlet side,  $E_2$  changes by  $4.4 \times 10^{-9}$  (EE) and  $2.0 \times 10^{-9}$  (II),  $E_1$  varies by 51% and 20% with the period of the first bending mode (10 ms), and the impulse residual is of the order of the impulse itself: the relative residual has the median 0.95 (EE) and 0.67 (II) and the maximum 3.4 and 4.8; the net residual is in the bending direction  $y$ , up to  $4.8 \times 10^{-2}$  and  $4.9 \times 10^{-2}$  Ns per window, and sums to  $-29$  and  $-17$  Ns over the 10 ms, while the  $x$  and  $z$  components stay below  $5 \times 10^{-12}$  Ns. The earlier fine-side runs with the former absolute floor changed  $E_2$  by  $7.1 \times 10^{-8}$  and  $2.4 \times 10^{-8}$ ; with the relative criterion the change is 40 and 12 times smaller, which identifies it as accumulated stopping error, as the derivation of Sec. 8.2 predicts. The  $E_1$  variation of the fine-side arrangement coincides with a nonzero impulse residual and the coarse-side arrangement has neither; whether the impulse residual causes the variation has not been tested separately. The mixed pairs of the earlier runs, IE with the coarse implicit side as the Dirichlet side ( $E_2$  within  $3.9 \times 10^{-12}$ ,  $E_1$  within 0.07%) and EI, whose coarse side is explicit and diverges as the Dirichlet side, are in Table 12.

**Subcycling on nonconforming meshes.** The coarse-step side must be the Dirichlet side for the flux structure above, and with identical integrators the finer mesh must be the Dirichlet side for the gain (Table 9). On the 1:0.5 beam (free part 6.35 mm, 160 elements; clamped part 12.7 mm, 20 elements) two assignments were run. In (A) the clamped (coarse-mesh) side takes  $10^{-6}$  s and is the Dirichlet side, and the free (fine-mesh) side takes  $2.5 \times 10^{-7}$  s; in (B) the free (fine-mesh) side takes  $10^{-6}$  s and is the Dirichlet side, and the clamped (coarse-mesh) side takes  $2.5 \times 10^{-7}$  s, so that the fine mesh is also the coarse-step side. The projector in use reproduces constants to  $10^{-15}$  in both, and  $\mathbf{B}\mathbf{1}$ ,  $\mathbf{B}^T\mathbf{1}$  match the boundary masses to  $10^{-16}$ . The gain of each assignment, linearized at a stop with the Neumann side’s end velocity per unit force ramped over its four substeps, is in the lower part of Table 11. In (A) with explicit sides the eigenvalues lie in  $[-7.9, -2.0]$  and the iteration diverges for every  $\theta \geq 0.25$ ; the run diverged already in the coupled initial acceleration at  $\theta = 0.5$  and at  $\theta = 0.25$  (kinetic energy of order  $10^{249}$  and  $10^{21}$  at  $t = 0$ ) and ended at the first stop. Assignment (B) conserves  $E_2$  to  $3.7 \times 10^{-12}$  (EE) and  $1.6 \times 10^{-12}$  (II) with the impulse residual at roundoff; assignment (A) with implicit sides, whose gain radius at  $\theta = 0.5$  is 0.55, conserves  $E_2$  to  $4.7 \times 10^{-13}$  in 44.5 iterations per stop against 18.8 for (B). The rule for subcycled pairs is therefore: the coarse-step side is the Dirichlet side; between two members with the same integrator the finer mesh should be that side, which may require giving the finer mesh the coarser step where the stability limit of an explicit member allows it; where it does not, the gain decides.

Table 11: Beam with one side subcycled 4:1, 10 ms,  $\theta = 0.5$ , relative tolerance  $10^{-12}$ , no absolute tolerance. D: the Dirichlet side; “coarse” and “fine” refer to the step.  $E_1(t)/E_1(0)$  at 1, 2, 5, and 10 ms;  $\delta E = \max_t |E/E(0) - 1|$  with the sign of the final value; drift: root mean square interface displacement jump at 10 ms (m); impulse: largest net residual per window in  $y$  (Ns) and the median relative residual; it.: Schwarz iterations per stop, mean/maximum (each iteration advances the fine side four substeps); wall time in seconds on four threads. The 1:0.5 (A) run with explicit sides ended at the first stop. Lower part: the spectral radius  $\rho$  of the relaxed iteration from the linearized gain at a stop.

| meshes | pair, D      | 1                                                                     | 2      | 5      | 10     | $\delta E_1$ | $\delta E_2$ | drift   | impulse $y$ , rel. | it.     | wall |
|--------|--------------|-----------------------------------------------------------------------|--------|--------|--------|--------------|--------------|---------|--------------------|---------|------|
| conf.  | EE, D coarse | 1.0005                                                                | 1.0009 | 1.0003 | 1.0006 | +1.1e-3      | -4.5e-13     | 3.1e-6  | 1.9e-15, 1.1e-13   | 6.0/7   | 139  |
| conf.  | EE, D fine   | 1.0432                                                                | 1.1376 | 1.4698 | 0.9938 | -5.1e-1      | +4.4e-9      | 4.2e-6  | 4.8e-2, 0.95       | 36.6/42 | 564  |
| conf.  | II, D coarse | 0.9968                                                                | 0.9982 | 0.9972 | 0.9971 | -3.8e-3      | +1.6e-12     | 3.1e-6  | 2.6e-14, 2.2e-13   | 10.1/15 | 560  |
| conf.  | II, D fine   | 1.0168                                                                | 1.0682 | 1.1977 | 0.9872 | -2.0e-1      | +2.0e-9      | 5.5e-14 | 4.9e-2, 0.67       | 35.7/40 | 1763 |
| 1:0.5  | EE, (A)      | diverged in the coupled initial acceleration; ended at the first stop |        |        |        |              |              |         |                    |         |      |
| 1:0.5  | EE, (B)      | 0.9994                                                                | 0.9995 | 0.9993 | 0.9993 | -8.9e-4      | +3.7e-12     | 1.5e-6  | 2.8e-17, 1.9e-16   | 26.9/31 | 315  |
| 1:0.5  | II, (A)      | 0.9989                                                                | 0.9992 | 0.9984 | 0.9993 | -1.9e-3      | -4.7e-13     | 2.2e-6  | 5.9e-14, 4.2e-13   | 44.5/54 | 2555 |
| 1:0.5  | II, (B)      | 1.0008                                                                | 1.0009 | 1.0008 | 1.0008 | +1.7e-3      | +1.6e-12     | 1.6e-6  | 2.4e-17, 1.9e-16   | 18.8/21 | 829  |

| meshes     | pair | Dirichlet side                    | $\rho(0.25)$ | $\rho(0.5)$ | $\rho(0.75)$ | $\rho(1)$ |
|------------|------|-----------------------------------|--------------|-------------|--------------|-----------|
| conforming | EE   | clamped, coarse step              | 0.50         | 0.008       | 0.50         | 1.00      |
| conforming | II   | clamped, coarse step              | 0.50         | 0.083       | 0.63         | 1.17      |
| 1:0.5      | EE   | (A) clamped, coarse mesh and step | 1.23         | 3.47        | 5.70         | 7.93      |
| 1:0.5      | II   | (A) clamped, coarse mesh and step | 0.26         | 0.55        | 1.32         | 2.09      |
| 1:0.5      | EE   | (B) free, fine mesh, coarse step  | 0.75         | 0.50        | 0.25         | 0.50      |
| 1:0.5      | II   | (B) free, fine mesh, coarse step  | 0.75         | 0.50        | 0.25         | 0.56      |

**The nested cylinders.** The cylinders of Sec. 7 (neohookean materials, central difference on both sides, stops of  $0.5 \mu\text{s}$ , nonconforming at 1:0.8) were run to 2 ms with the constrained exchange, velocity constraint, the inner (finer) side as the Dirichlet side,  $\theta = 0.5$ , and the tolerances of the example (relative  $10^{-6}$ , absolute  $10^{-4}$ ). The materials are nonlinear, so  $E_2$  is not defined and the question is whether  $E_1$  stays bounded. With the inner side as the Dirichlet side only  $\Pi_D$  enters

the exchange, and its partition-of-unity error is  $4 \times 10^{-15}$ ; the projector onto the outer side, which the exchange does not use, has an error of 0.72 at 2578 of its 2582 nodes, the quadrature error of  $\mathbf{B}$  on a curved, non-nested interface. The coupled initial acceleration converged in 14 iterations.

| coupling                          | 0.5 ms             | 1 ms    | 2 ms    | it. per stop | note              |
|-----------------------------------|--------------------|---------|---------|--------------|-------------------|
| monolithic                        | 1.000              | 1.000   | 1.000   | –            | 0.01% to 5 ms     |
| present DN, Aitken                | 1.00               | 1.00    | 1.01    | 2            | +3, –1.3% to 5 ms |
| paired impedance                  | 0.79               | 0.54    | 0.24    | 2            |                   |
| constrained, abs. tol. $10^{-4}$  | 1.00095            | 1.00389 | 1.02073 | 1            | 5943 s, 8 threads |
| constrained, abs. tol. $10^{-12}$ | 0.999976 at 0.1 ms |         |         | 8.8 (max 10) | 1218 s            |

With the example’s absolute tolerance the constrained iteration stopped after one iteration at every stop: the absolute branch of the stopping rule accepts a root mean square velocity jump below the absolute tolerance divided by the step, here  $10^{-4} \text{ m} / 5 \times 10^{-7} \text{ s} = 200 \text{ m/s}$ , while the measured jump was 0.27 to 0.46 m/s (0.6 to 1.4% relative). The run is therefore the unconverged staggered exchange, and its  $E_1$  grows by 2.1% in 2 ms at an increasing rate. With the absolute tolerance  $10^{-12}$  the iteration contracts (no stall), needs 8.8 iterations per stop, and leaves a relative velocity jump of  $4.6 \times 10^{-7}$ ; over the 0.1 ms run  $E_1$  stays within  $2.4 \times 10^{-5}$ , against  $9 \times 10^{-5}$  for the one-iteration run at the same time. A converged run to 2 ms would take about 7 hours on 8 threads and was not made, so boundedness under convergence is shown only to 0.1 ms.

**The stopping floor.** The constrained criterion first accepted a root mean square jump of the constrained velocity below the controller’s absolute tolerance divided by the step; the absolute tolerance is a displacement, and on the cylinders the floor was 200 m/s. The criterion is now relative only, unless `constraint absolute tolerance` gives a tolerance in the units of the constrained quantity (default 0). The beam runs below use the relative tolerance  $10^{-12}$  and no absolute tolerance.

**Direct interface solve.** For two central difference subdomains at equal steps the iteration can be replaced by one solve. Let  $\lambda$  be the force on the interface rows of the Dirichlet side  $D$ , so that the Neumann side  $N$  receives  $-\Pi_D^T \lambda$ , and let  $\mathbf{m}_D$  and  $\mathbf{m}_N$  be the lumped masses of the interface rows. Central difference computes  $\mathbf{u}_{n+1}$  before the acceleration, and the internal force at  $\mathbf{u}_{n+1}$  does not depend on  $\lambda$ . Advancing both sides with free interface rows and no interface force gives the velocities  $\mathbf{v}^{\text{free}}$ ; the interface force adds  $\gamma \Delta t \mathbf{m}_D^{-1} \lambda$  to the velocity of  $D$  and  $-\gamma \Delta t \mathbf{m}_N^{-1} \Pi_D^T \lambda$  to that of  $N$ , and the velocity constraint  $\mathbf{v}_D = \Pi_D \mathbf{v}_N$  at the end of the step gives, per Cartesian component,

$$\gamma \Delta t (\mathbf{m}_D^{-1} + \Pi_D \mathbf{m}_N^{-1} \Pi_D^T) \lambda = \Pi_D \mathbf{v}_N^{\text{free}} - \mathbf{v}_D^{\text{free}}. \quad (25)$$

The matrix is symmetric positive definite, of the size of the Dirichlet interface, and constant while the lumped masses and  $\Pi_D$  are; rows held by other Dirichlet conditions take a zero inverse mass. It is  $\mathbf{H}$  of (20) for two lumped-mass explicit sides, written on the Dirichlet trace. It is factored once (dense Cholesky: 3920 unknowns per component on the cylinders, three factorizations in 0.7 s,  $3 \times 123 \text{ MB}$ ). Each stop advances both sides, solves (25), and corrects the interface accelerations and velocities. Because the internal force is evaluated at the predictor displacement, the solve is exact for nonlinear materials as well. The same matrix gives the interface force at  $t = 0$  from the acceleration constraint  $\mathbf{a}_D = \Pi_D \mathbf{a}_N$  (`interface solve: direct`).

| case                        | exchange   | $\delta E_1$ | $\delta E_2$ | drift (m) | it.     | wall (s) | tolerance       |
|-----------------------------|------------|--------------|--------------|-----------|---------|----------|-----------------|
| beam, conforming, EE        | direct     | 1.5e-14      | 4.4e-14      | 1.6e-17   | –       | 39       | –               |
|                             | iterated   | 1.2e-14      | 3.6e-14      | 4.8e-17   | 1.0/1   | 45       | rel. $10^{-12}$ |
| beam, 1:0.5, EE             | direct     | 7.8e-15      | 4.5e-14      | 1.8e-17   | –       | 39       | –               |
|                             | iterated   | 2.2e-12      | 4.4e-12      | 4.5e-15   | 26.8/30 | 176      | rel. $10^{-12}$ |
| beam, 1:0.75, EE            | direct     | 1.2e-14      | 6.0e-14      | 2.7e-17   | –       | 39       | –               |
|                             | iterated   | 1.6e-12      | 4.6e-12      | 2.3e-15   | 18.1/20 | 164      | rel. $10^{-12}$ |
| linear cylinders, $-5$ m/s  | monolithic | 1.3e-13      | 6.2e-12      | –         | –       | 371      | –               |
|                             | direct     | 9.9e-14      | 3.2e-12      | 6.0e-19   | –       | 735      | –               |
| linear cylinders, $-35$ m/s | monolithic | 1.2e-14      | 4.8e-13      | –         | –       | 359      | –               |
|                             | direct     | 1.1e-14      | 4.3e-13      | 4.3e-18   | –       | 731      | –               |

Beam runs: 10 ms,  $10^{-6}$  s; cylinders: 1 ms,  $0.5 \mu\text{s}$ , the cylinders on 8 threads.  $\delta E$  is the largest  $|E/E(0) - 1|$ , drift the root mean square interface displacement jump at the end. The direct solve conserves both energies to roundoff, at a lower cost than the iteration on the nonconforming meshes, and on the cylinders the relative interface velocity jump at the end of every stop is at most  $7 \times 10^{-16}$ . On the linear cylinders the monolithic  $E_2$  varies by  $6.2 \times 10^{-12}$  at  $-5$  m/s and  $4.8 \times 10^{-13}$  at  $-35$  m/s, the decomposed one by  $3.2 \times 10^{-12}$  and  $4.3 \times 10^{-13}$ : the level depends on the amplitude because the deformation gradient is formed from the current coordinates  $\mathbf{X} + \mathbf{u}$ , which loses digits in proportion to  $|\mathbf{X}|/|\mathbf{u}|$ . The per-step increments of  $E_2$  have a standard deviation of  $1.1 \times 10^{-13}$  and a lag-one autocorrelation of  $-0.31$ , without a significant mean: roundoff in the evaluation and its accumulation, not a systematic change. On the neohookean cylinders ( $-35$  m/s, 2 ms) the direct solve completes in 1416 s with  $E_1/E_1(0) = 0.999972$ ,  $0.999963$ , and  $0.999936$  at 0.5, 1, and 2 ms, within the 0.01% that the monolithic reference keeps to 5 ms, a relative velocity jump of at most  $7 \times 10^{-16}$ , and a drift of  $6 \times 10^{-18}$  m; the iterated run with the former floor reached  $+2.1\%$  at 2 ms.

**Comparison on the beam.** Table 12 sets the monolithic reference, the present Dirichlet–Neumann exchange (`constrained: false`, free side Dirichlet,  $\theta = 0.5$ , tolerances of the examples: relative  $10^{-8}$ , absolute  $2.54 \times 10^{-8}$ , at most 128 iterations; Newton with conjugate gradients, relative  $10^{-10}$ , absolute  $2.54 \times 10^{-8}$ ), the paired impedance coupling (defaults: harmonic-mean impedance, Robin parameter 0;  $\theta = 0.5$  with an explicit member, recursive Aitken for II; the same tolerances), and the constrained exchange (tolerance  $10^{-12}$ , absolute  $10^{-15}$ , at most 256 iterations, direct linear solver, coupled initial acceleration,  $\theta = 0.5$ ) side by side, all at the stop  $10^{-6}$  s and to 10 ms. The examples use  $5 \times 10^{-7}$  s; the present exchange is run here at  $10^{-6}$  s.

**Paired impedance or constrained exchange.** On every case of Table 12 the constrained exchange conserves at least as well as the paired impedance coupling and needs comparable iteration counts: on conforming meshes the paired coupling is within  $3.5 \times 10^{-3}$  for II and the mixed pairs and loses 0.55% for EE, while the constrained exchange is within the tolerance; at 1:0.5 the paired coupling dissipates 57 to 58% of  $E_1$ , which the dashpot takes from the jump it cannot close, and at 1:0.75 it ends by element inversion at 3.5 and 5.0 ms, while the constrained exchange conserves to  $10^{-10}$  at both ratios; with subcycling it loses 9 and 22% against 0.11 and 0.38%. The impedance coupling is therefore not the solver for any case measured on the beam. It remains of interest only where the constrained exchange has no Dirichlet side to offer, a mixed pair with subcycling whose coarse member is explicit, and for interfaces whose two sides cannot be ordered by the gain (24); reformulating it as a solver of the constrained problem, with the impedance as a preconditioner of the interface iteration rather than as part of the transmission condition, is the form in which it could serve there. No code was removed.

Table 12: Cantilever, 10 ms.  $E_1(t)/E_1(0)$  at 1, 2, 5, and 10 ms;  $\delta E = \max_t |E(t)/E(0) - 1|$  with the sign of  $E(10 \text{ ms})/E(0) - 1$  (or of the last value reached); drift is the root mean square one-sided interface displacement jump at 10 ms in m; it. is the mean and maximum number of Schwarz iterations per stop; wall time in seconds on four threads; the last column gives completion or the time and cause of the end of the run.  $v$  and  $u$ : velocity and displacement constraint; D: the Dirichlet side.

| coupling                                                                              | 1      | 2       | 5      | 10     | $\delta E_1$ | $\delta E_2$ | drift   | it.     | wall | completed        |
|---------------------------------------------------------------------------------------|--------|---------|--------|--------|--------------|--------------|---------|---------|------|------------------|
| <i>conforming meshes (8.47 mm), equal steps</i>                                       |        |         |        |        |              |              |         |         |      |                  |
| monolithic I                                                                          | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -2.4e-14     | -7.7e-14     | -       | 1.0/1   | 71   | yes              |
| monolithic E                                                                          | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -1.5e-14     | -2.9e-14     | -       | 1.0/1   | 40   | yes              |
| II present DN                                                                         | -      | -       | -      | -      | +6.6e0       | +2.2e2       | -       | 15.6/21 | -    | inverted 0.57 ms |
| II impedance                                                                          | 1.0000 | 1.0000  | 1.0000 | 1.0000 | +1.0e-7      | -1.7e-6      | 1.3e-11 | 6.5/8   | 438  | yes              |
| II constr. $v$                                                                        | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -2.8e-11     | -7.0e-11     | 5.8e-15 | 7.0/8   | 211  | yes              |
| II constr. $u$                                                                        | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -2.2e-11     | -5.3e-11     | 2.4e-15 | 7.1/8   | 195  | yes              |
| IE present DN                                                                         | 1.0025 | 1.0018  | 1.0066 | 1.0052 | +2.9e-2      | +2.3e-1      | 4.8e-9  | 13.1/14 | 303  | yes              |
| IE impedance                                                                          | 0.9985 | 0.9990  | 0.9986 | 0.9973 | -3.5e-3      | -7.7e-2      | 5.1e-6  | 22.6/27 | 645  | yes              |
| IE constr. $v$ , D implicit                                                           | 0.9990 | 0.9997  | 0.9999 | 0.9996 | -1.2e-3      | -6.3e-12     | 5.2e-6  | 29.0/32 | 380  | yes              |
| EI present DN                                                                         | 1.2185 | 5.2604  | -      | -      | +1.3e1       | +4.6e1       | -       | 15.3/18 | -    | inverted 2.39 ms |
| EI impedance                                                                          | 1.0031 | 1.0011  | 1.0013 | 1.0011 | +3.3e-3      | -5.1e-2      | 2.8e-6  | 23.0/27 | 718  | yes              |
| EI constr. $v$ , D implicit                                                           | 1.0033 | 1.0016  | 1.0024 | 1.0034 | +3.7e-3      | +4.2e-10     | 2.9e-6  | 27.1/28 | 377  | yes              |
| EE present DN                                                                         | 1.0037 | 1.0019  | 1.0009 | 1.0238 | +4.3e-2      | +5.4e-2      | 7.6e-9  | 13.1/14 | 114  | yes              |
| EE impedance                                                                          | 0.9997 | 0.9994  | 0.9974 | 0.9945 | -5.5e-3      | -6.6e-2      | 1.2e-5  | 31.0/86 | 219  | yes              |
| EE constr. $v$                                                                        | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -1.2e-14     | -3.6e-14     | 4.8e-17 | 1.0/1   | 51   | yes              |
| <i>free part 6.35 mm; 1:0.5 and 1:0.75 coarsen the clamped part; equal steps</i>      |        |         |        |        |              |              |         |         |      |                  |
| monolithic I                                                                          | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -6.1e-14     | -1.1e-13     | -       | 1.0/1   | 93   | yes              |
| monolithic E                                                                          | 1.0000 | 1.0000  | 1.0000 | 1.0000 | +1.0e-14     | -3.5e-14     | -       | 1.0/1   | 45   | yes              |
| 1:0.5 II present DN                                                                   | -      | -       | -      | -      | +3.0e1       | +3.4e2       | -       | 14.8/19 | -    | inverted 0.69 ms |
| 1:0.5 II impedance                                                                    | 0.8668 | 0.7793  | 0.6058 | 0.4317 | -5.7e-1      | -7.0e-1      | 3.8e-5  | 6.0/7   | 403  | yes              |
| 1:0.5 II constr. $v$                                                                  | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -1.2e-10     | -1.8e-11     | 3.7e-14 | 17.2/18 | 448  | yes              |
| 1:0.5 II constr. $u$                                                                  | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -1.2e-10     | +4.3e-11     | 7.6e-15 | 17.3/19 | 463  | yes              |
| 1:0.5 EE present DN                                                                   | 1.0073 | 1.0017  | 1.0005 | 1.0083 | +2.8e-2      | +1.9e-2      | 5.7e-9  | 12.9/15 | 108  | yes              |
| 1:0.5 EE impedance                                                                    | 0.8953 | 0.7742  | 0.5963 | 0.4157 | -5.8e-1      | -7.7e-1      | 1.5e-5  | 21.2/27 | 158  | yes              |
| 1:0.5 EE constr. $v$                                                                  | 1.0000 | 1.0000  | 1.0000 | 1.0000 | +3.5e-11     | +6.9e-11     | 1.0e-13 | 23.7/26 | 173  | yes              |
| 1:0.75 II present DN                                                                  | -      | -       | -      | -      | +9.8e0       | +2.5e2       | -       | 15.0/20 | -    | inverted 0.38 ms |
| 1:0.75 II impedance                                                                   | 0.9573 | 0.9321  | 0.8717 | -      | -1.3e-1      | +3.0e4       | -       | 6.8/8   | -    | inverted 5.02 ms |
| 1:0.75 II constr. $v$                                                                 | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -1.7e-11     | -5.2e-11     | 8.5e-15 | 13.5/14 | 425  | yes              |
| 1:0.75 II constr. $u$                                                                 | 1.0000 | 1.0000  | 1.0000 | 1.0000 | -4.2e-11     | +3.2e-10     | 3.5e-15 | 13.1/15 | 391  | yes              |
| 1:0.75 EE present DN                                                                  | 1.0059 | 1.0056  | 1.0026 | 1.0020 | +2.6e-2      | +2.3e-2      | 1.0e-8  | 12.9/14 | 131  | yes              |
| 1:0.75 EE impedance                                                                   | 0.9489 | -1.3453 | -      | -      | -2.4e3       | +3.3e5       | -       | 35.6/51 | -    | inverted 3.52 ms |
| 1:0.75 EE constr. $v$                                                                 | 1.0000 | 1.0000  | 1.0000 | 1.0000 | +2.2e-11     | +6.1e-11     | 1.6e-14 | 16.2/17 | 163  | yes              |
| <i>conforming meshes, free part subcycled 4:1 (<math>2.5 \times 10^{-7}</math> s)</i> |        |         |        |        |              |              |         |         |      |                  |
| EE present DN                                                                         | 1.0017 | 1.0012  | 1.0053 | 1.0089 | +3.5e-2      | +5.4e-2      | 8.6e-9  | 13.0/14 | 219  | yes              |
| EE impedance                                                                          | 0.9907 | 0.9825  | 0.9530 | 0.9100 | -9.0e-2      | -1.6e-1      | 9.0e-6  | 32.0/62 | 424  | yes              |
| EE constr. $v$ , D fine                                                               | 1.0432 | 1.1376  | 1.4698 | 0.9938 | -5.1e-1      | +7.1e-8      | 4.2e-6  | 32.6/34 | 518  | yes              |
| EE constr. $v$ , D coarse                                                             | 1.0005 | 1.0009  | 1.0003 | 1.0006 | +1.1e-3      | -4.5e-13     | 3.1e-6  | 6.0/7   | 131  | yes              |
| II present DN                                                                         | -      | -       | -      | -      | +6.4e0       | +2.0e2       | -       | 15.6/21 | -    | inverted 0.36 ms |
| II impedance                                                                          | 0.9717 | 0.9427  | 0.8745 | 0.7830 | -2.2e-1      | -4.7e-1      | 4.4e-6  | 6.0/6   | 771  | yes              |
| II constr. $v$ , D fine                                                               | 1.0168 | 1.0682  | 1.1977 | 0.9872 | -2.0e-1      | +2.4e-8      | 4.2e-14 | 32.2/34 | 1335 | yes              |
| II constr. $v$ , D coarse                                                             | 0.9968 | 0.9982  | 0.9972 | 0.9971 | -3.8e-3      | +1.3e-12     | 3.1e-6  | 10.1/15 | 636  | yes              |
| IE constr. $v$ , D coarse (I)                                                         | 0.9994 | 0.9998  | 0.9999 | 0.9997 | -7.4e-4      | -3.9e-12     | 1.5e-6  | 28.9/32 | 643  | yes              |
| EI constr. $v$ , D fine (I)                                                           | 1.0167 | 1.0477  | 1.1465 | 0.9942 | -1.7e-1      | +3.4e-8      | 5.1e-6  | 33.4/35 | 1326 | yes              |
| EI constr. $v$ , D coarse (E)                                                         | -      | -       | -      | -      | NaN          | NaN          | -       | 1/1     | -    | inverted 0.00 ms |

## 9 Conclusions

### 9.1 Findings

1. Dirichlet transmission on nonconforming meshes injects energy on both benchmarks, on the overlap decomposition and on the implicit nonoverlap Dirichlet–Neumann exchange: exponential growth ending in element inversion, at comparable rates on a flexural beam and on a shock-loaded nested assembly (Secs. 6.2, 6.6, 7).
2. The classical Robin–Robin condition is not a remedy. It reflects every wave ( $|R| \equiv 1$ ), has no term that damps the interface mode, and pumps energy on both benchmarks and both decompositions, usually faster than Dirichlet transmission, at either Robin parameter; with per-side transfer its converged solution depends on  $\alpha$  (Secs. 6.7, 7).
3. On an overlap boundary the impedance condition is consistent with the monodomain solution only with the partner traction, and with the consistent traction it is exact on conforming meshes (0.08% at 1 ms, no parameters). On nonconforming meshes its interface work is sign-indefinite in principle (two surfaces, non-adjoint transfers) and in practice (injection up to +275%, sign changes between adjacent ratios, 74 of 108 runs failing by 10 ms), and no construction at the interface changes that, because the discrepancy is the difference of the two grids’ dispersion relations, which both grids represent (Secs. 6.1–6.3). On the cylinders it fails sooner than Dirichlet coupling.
4. Energy consistency requires one interface traction acting through adjoint transfers built from one cross-mass matrix; the interface power is then the sign-definite dissipation (14) on the interface jump. Only the nonoverlap decomposition, with its shared interface, admits this construction (Sec. 4).
5. The adjoint-paired nonoverlap coupling never injects energy at any ratio, pairing, or benchmark. On the beam it conserves to 0.3% at 1 ms and to 0.0% at 10 ms on conforming and 1:0.9 meshes with an implicit member; at strong contrast it dissipates the unrepresentable jump at a persistent rate (40 to 85% by 10 ms); the non-nested ratios 1:0.7 to 1:0.8 accumulate interface-local high-frequency content until an element inverts at 3 to 5 ms; and explicit–explicit pairs end bounded but noisy, which subcycling the explicit member removes (Sec. 6.4).
6. On the cylinders the paired coupling is stable but loses half the energy within 1 ms, because the iteration stops on displacement while the interface velocity jump, which the dashpot damps, stalls at a finite value; more iterations halve the loss, and no relaxation nearly removes it before the iteration diverges (Sec. 7).
7. Different time steps per subdomain use piecewise-linear interpolation of the partner trajectory, whose dissipation of order  $\Delta t$  is the energy loss intrinsic to subcycling; exact cancellation of the subcycled interface work is incompatible with a partitioned exchange (Sec. 5.3).
8. The relaxation state must be keyed per pair and per substep slot; Aitken acceleration applies to single-slot stops only, reduces the iteration count tenfold on the implicit beam pairs, and diverges in every form on the explicit cylinders pair, where a fixed factor of 0.5 converges in 2 to 3 iterations (Sec. 5.4).
9. With explicit subdomains on both sides the plain Dirichlet–Neumann exchange conserved on both benchmarks (within 1% on the beam, 3% on the cylinders), while with implicit subdomains it is the fastest coupling to fail. Its interface force misses the interface inertia:

the relative force residual against the d'Alembert reaction has the median 0.83 (explicit) and 1.47 (implicit) (Sec. 8.5).

10. The constrained Dirichlet–Neumann exchange (d'Alembert reaction, transfers from one cross matrix, one imposed quantity with the other two from the receiver's Newmark relations, stopping on the constraint and the force residual, coupled initial acceleration) conserves the matched energy on the beam over 10 ms to between  $4 \times 10^{-14}$  and  $4 \times 10^{-10}$  at a tolerance of  $10^{-12}$ , for II, IE, EI, and EE, on conforming meshes and at the ratios 1:0.5 and 1:0.75; at 1:0.75, where the paired impedance coupling inverted elements, it completes (Sec. 8.5).
11. The Dirichlet role decides convergence: with the explicit member as the Dirichlet side the gain of the iteration has eigenvalues down to  $-13.3$  and every relaxation factor from 0.5 to 1 diverges; with the implicit member they lie in  $[-0.61, -0.075]$ . For identical integrators on conforming meshes the eigenvalues are near  $-1$ :  $\theta = 0.5$  is optimal and  $\theta = 1$  does not converge.
12. With a 4:1 step ratio the constrained exchange with the coarse-step side as the Dirichlet side keeps  $E_2$  within  $4 \times 10^{-12}$  and  $E_1$  within 0.4% over 10 ms, without a direct interface solve (Sec. 8.2); with the fine-step side as the Dirichlet side  $E_1$  varies by up to 51% and the interface impulse residual is of the order of the impulse. The present paired impedance coupling loses 9 (EE) and 22% (II) on the same case (Sec. 8.5).
13. For two explicit subdomains at equal steps the direct interface solve (25) conserves  $E_1$  and  $E_2$  to roundoff on the beam at every mesh ratio and on the linear cylinders at  $-5$  and  $-35$  m/s, with no iteration, and keeps  $E_1$  of the neohookean cylinders within  $6.4 \times 10^{-5}$  over 2 ms.
14. On the beam the constrained exchange conserves at least as well as the paired impedance coupling in every case measured, including the nonconforming ratios where the paired coupling dissipates 58% (1:0.5) or inverts an element (1:0.75).

## 9.2 Recommendations

- **Energy-critical dynamic coupling:** use the constrained Dirichlet–Neumann exchange (`constrained: true`, velocity constraint,  $\theta = 0.5$ ), with the implicit member, or with identical integrators the finer side, as the Dirichlet side at equal steps, and the coarse side as the Dirichlet side with different steps. The adjoint-paired impedance coupling is retained in the code; on the beam it conserved no case better than the constrained exchange (Sec. 8.5).
- **Overlap decompositions:** conforming meshes with implicit–implicit integration conserve at any horizon; every other overlap configuration is for short transients only.
- **Robin–Robin:** for quasi-statics and comparison studies only.
- **All-explicit interfaces:** subcycle the finer side, or accept the bounded, noisy state; central difference removes no grid-frequency content at any stable step.
- **Parameters:** the impedance is material data (harmonic mean per pair for nonoverlap, partner material with P/S split for overlap); no scaling. The Robin parameter must be one value per interface under pairing and affects only the iteration when the jump can close; on non-conforming meshes its effect on the converged solution is unmeasured for the paired coupling, and the runs in this note use  $\alpha = 0$  or  $2 \times 10^9$  N/m<sup>3</sup>.
- **Path to conservation:** at equal time steps use the constrained Dirichlet–Neumann exchange (`constrained: true`) with the implicit member, or the finer side, as the Dirichlet side and a

fixed relaxation factor chosen from the gain (24) (0.5 for identical integrators); the remaining changes of Sec. 8.4 (direct interface solve, different time steps) come next.

- **Diagnostics:** use the blended energy output with the staggered metric for explicit members, and the channel-resolved interface-work instrumentation as the acceptance metric for any change to the transmission machinery; a small net drift can be the difference of two large channels.

### 9.3 Open questions

- (i) The stalled interface jump on the cylinders: a stopping criterion that refuses a stalled velocity jump, relaxation factors between 0.5 and 1, and subcycling of the explicit side are the candidates for recovering the energy the paired coupling now dissipates.
- (ii) The dependence of the converged paired solution on  $\alpha$  and on  $\theta$ : the fixed point differs by about  $4 \times 10^{-5}$  relative displacement between  $\theta = 0.5$  and 1 at a  $10^{-13}$  tolerance on the 1:0.5 beam, evidence of a neutral mode of the exchange that the displacement criterion does not measure; an  $\alpha$  sweep of the paired coupling on nonconforming meshes has not been run.
- (iii) Why the Dirichlet–Neumann exchange is stable between explicit subdomains and unstable between implicit ones. Answered by the first reference experiment: with the departures of Sec. 8.3 removed, the implicit pair conserves  $E_1$  and  $E_2$  to  $7 \times 10^{-11}$  over 10 ms, where the present exchange inverts an element at 0.46 ms; the present exchange misses an interface force of the size of the reaction itself (median relative residual 1.47 implicit, 0.83 explicit). Which departure alone accounts for the difference between the two integrators was not isolated.
- (iv) The non-nested mesh-ratio failure window of the paired transfer at long horizons.
- (v) Higher-order multirate coupling ( $r \geq 2$  in the sense of Connors et al. [5]) and a direct interface solve over a subcycled window. Exact cancellation of the subcycled interface work itself is obtained by the iterated exchange with the coarse-step side as the Dirichlet side (Sec. 8.2).
- (vi) Under subcycling with the fine-step side as the Dirichlet side,  $E_2$  is conserved up to accumulated stopping error ( $4.4 \times 10^{-9}$  EE and  $2.0 \times 10^{-9}$  II over 10 ms at the relative tolerance  $10^{-12}$ ), but  $E_1$  varies by 51% and 20% at the first bending period, and the interface impulse residual is of the order of the impulse (median relative residual 0.95 and 0.67, net  $y$  residual summing to  $-29$  and  $-17$  N s), against roundoff for the coarse-step Dirichlet side. Whether the impulse residual causes the  $E_1$  variation has not been tested separately.
- (vii) The absolute branch of the constrained stopping rule divided the absolute tolerance by the step to obtain a velocity floor of 200 m/s on the cylinders. The floor is removed and the direct solve converges the explicit exchange exactly (Sec. 8.5); a converged iterated cylinder run to 2 ms has not been made.
- (viii) Mesh-ratio and overlap-width families on the cylinders, and implicit versions of its subdomains, to test whether the beam conclusions transfer.

### Data provenance

Correction-sequence study and instrumentation: `rigel:~/test/Norma/cantilever-energy`. Mesh-ratio sweep, 10 ms horizon, controller-step sweep, and run harness: `sirius:~/cantilever-energy-study/`. The 2026-09 reruns, the Robin–Robin runs, and the cylinders runs: `rigel:~/test/Norma/beam/` (runs, runs-r090, runs-r050) and `rigel:~/test/Norma/`

cylinders/. Slides: `Norma-Energy-Stability-Beam-Cylinders-2026-09.pptx`. Regression tests: 104 (overlap impedance energy and consistent-traction patch), 105 (tensor impedance cross-scaling), 106 (variational transfer projectors and content-absorption dissipativity), 110 (adjoint pairing, windowed equivalence, keyword guards), 125 (explicit stable step with subcycling).

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